

# Macprudential Regulation, Quantitative Easing, and Bank Lending

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We show that widely used macroprudential regulations that rely on historical cost accounting (HCA) to insulate banks' balance sheets from financial market volatility significantly affect the transmission of monetary policy onto bank lending. Using detailed supervisory data from Italian banks, we find that HCA mutes the transmission of quantitative easing and other monetary policies that affect the long end of the yield curve, weakening the effectiveness of interventions aimed at reducing firm credit constraints. We suggest alternative policies that have the benefits of HCA but allow monetary policy to pass through. (*JEL* E52, G21, M48)

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Sovereign securities are a major part of banks' asset holdings, and volatility in sovereign asset prices can lead to financial instability and credit crunches.

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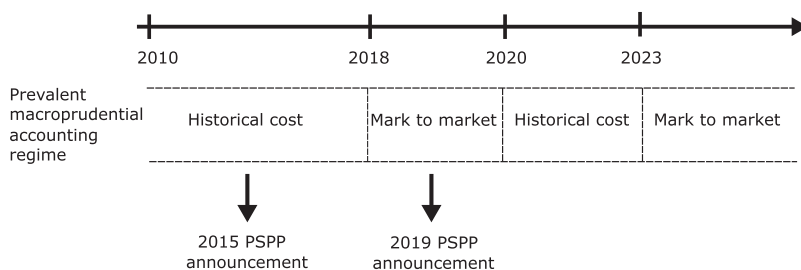
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**Figure 1****Macroprudential accounting regime of sovereign securities and QE announcements**

The figure shows the evolution of some key aspects of the European Union macroprudential accounting regime that banks use to value their holdings of sovereign securities when computing regulatory net worth (see Section 2 for more details) and the two main announcements of the PSPP, which is the ECB's largest QE program.

Recent examples of this phenomenon include the sovereign-bank “diabolic loop” in Europe in the early 2010s, the collapse of Silicon Valley Bank (SVB) and First Republic Bank, and the distress at some regional U.S. banks in 2023.<sup>1</sup> To help contain the risks from this volatility, regulators use accounting-based macroprudential policies, a crucial part of the broader macro-stabilization tool kit.<sup>2</sup> During the European sovereign-bank crisis of the early 2010s, most sovereign assets held by European banks were valued using historical cost accounting (HCA) to shield banks’ regulatory capital and lending capacity from sovereign bond volatility. When the crisis ebbed in 2018, regulators used mark-to-market accounting (MMA) more widely, allowing sovereign price changes to pass through onto banks’ balance sheets (Figure 1).<sup>3</sup> Alongside these accounting-based macroprudential policies, central banks also use yield curve management policies, such as forward guidance and quantitative easing (QE), to stabilize economies. However, little is known about how these macroprudential tools mediate the transmission of monetary policy onto the availability of bank credit to firms. Our paper addresses this gap.

Our main focus is on the European Central Bank’s largest QE program—the Public Sector Purchase Programme (PSPP)—together with the changes in macroprudential regulation shown in Figure 1. In particular, we study the

<sup>1</sup> See Brunnermeier et al. (2016), Gennaioli, Martin, and Rossi (2014), and Jiang et al. (2024). SVB and First Republic Bank experienced losses on their securities triggered by increases in interest rates, and a run precipitated their failure.

<sup>2</sup> The active use of accounting rules in banking regulation began in the early 1990s in the wake of the savings and loans crisis. Mark-to-market accounting was intended to prevent banks from hiding bad assets and “gains trading” their good assets, and historical cost accounting was intended to reduce the volatility of equity and earnings (Ellul et al. 2015; Greenspan 1990; Shaffer 1992).

<sup>3</sup> Accounting-based macroprudential policies are employed in other jurisdictions too. In the United States, HCA is used by most banks, MMA is prevalent only for larger “advanced approaches” banks (Fuster and Vickery 2018; Kim, Kim, and Ryan 2019), and the prevalence of HCA increased when interest rates rose in both 1994–1995 and 2022. In particular, in 2022, regulators allowed six large banks to move \$500 billion in assets from the MMA to HCA classification, as reported by the *Wall Street Journal* (Weil 2023). In Australia, MMA is prevalent (Lonsdale 2023).

introduction of the PSPP in January 2015, when HCA was used to value most sovereign securities, and the relaunch of the PSPP in 2019, when MMA was more prevalent. These policy changes, along with credit register and granular supervisory data, allow us to make progress on causal inference and identify the pathways through which macroprudential tools shape the transmission of QE onto bank lending. We then go beyond this specific setting and broaden the analysis to understand the interaction between macroprudential accounting tools and more general monetary policies that affect the yield curve, such as forward guidance and other asset purchase programs. Throughout, we focus on banks and firms in Italy, the largest country affected by the sovereign-bank “diabolic loop.”

The impact of macroprudential rules, such as HCA, on the transmission of monetary policy interventions like QE is an empirical question and depends on the mechanism through which QE works. Notably, if the bank recapitalization or portfolio rebalancing channels are salient, HCA might inhibit the transmission of QE onto bank lending. According to the recapitalization channel, an increase in asset prices triggered by QE has a positive impact on banks’ net worth and lending capacity. In the case of the portfolio rebalancing channel, banks sell their appreciated assets in order to rebalance their portfolio toward riskier, higher-return assets, such as loans. For either channel, the widespread use of HCA would limit the pass-through of any QE-related asset price increase onto the balance sheet of banks, attenuating the impact on bank lending. But the macroprudential accounting regime could be irrelevant for the transmission of QE and other monetary policies if these interventions operate in other ways, such as by easing liquidity conditions for banks or increasing the market price of bank equity.

There are significant challenges in identifying whether the macroprudential accounting regime mediates the transmission of QE onto bank loan supply. First, both macroprudential and monetary policy respond endogenously to current and expected economic conditions (Nakamura and Steinsson 2018).<sup>4</sup> Second, QE is often implemented alongside other policies that can also affect bank lending. And third, accounting-based macroprudential policies are highly targeted tools, and changes in the price of an asset can affect a bank’s net worth very differently depending on where the asset is recorded on the balance sheet.<sup>5</sup> We address these issues by using monthly supervisory Italian bank balance sheet data and monthly firm-bank-level data from the Italian credit register. The relatively high frequency of monthly data reduces the risk of

<sup>4</sup> In our setting, the risk of deflation and weak credit supply in the aftermath of the European sovereign crisis spurred the 2015 PSPP announcement, while improved economic conditions fostered the 2018 change in macroprudential policy toward MMA.

<sup>5</sup> For example, different macroprudential accounting rules apply for an asset recorded in the trading book on a bank’s balance sheet relative to the same asset recorded as held to maturity. As a result, analyses based on a broad aggregation of a bank’s asset holdings suffer from considerable measurement error and cannot identify the relevant transmission channels.

contamination from other policies. The use of credit register data allows us to nonparametrically control for loan demand (Khwaja and Mian 2008). And granular supervisory data on banks' asset holdings, including the specific location on banks' balance sheets and their accounting treatment, allow us to precisely measure the pass-through of QE on each bank.

We find that HCA mutes the impact of QE on bank lending. In both 2015 and 2019, only banks exposed to the PSPP through their holdings of MMA sovereign assets increased lending in response to the PSPP. The economic impact of the PSPP is vastly different across the two periods, consistent with the variation in the use of HCA. In 2015, HCA was widely used to protect the banking system, and for sovereign securities, MMA covered only those classified in the trading book. Because few banks had significant trading book holdings, the 2015 PSPP increased lending by approximately €214 million according to a back-of-the-envelope calculation. The lending response in 2019 is about 56 times larger, at about €12 billion, reflecting the fact that MMA had become more prevalent, increasing the size of banks' exposure to PSPP-eligible asset purchases. We also find that much of this additional lending went to less risky firms, and banks generally tried to reduce their risk exposure after the PSPP with riskier banks derisking the most.

A battery of robustness tests rules out concerns related to anticipatory biases, selection into exposure, and seasonality, and show that our results are robust when we control for several banks' characteristics as well as banks' exposure to other central bank interventions. We also show that interest rates declined by about 100 and 60 basis points (bps) for loans originated by exposed banks in 2015 and 2019, respectively, relative to less exposed ones. The decrease in rates, along with the increase in loan volume, implies that we are identifying changes in supply as opposed to movements in credit demand.<sup>6</sup>

We then turn our attention to identifying the mechanism behind the transmission of QE to discern the role played by macroprudential accounting rules. In both 2015 and 2019, the recapitalization channel appears to be the key mechanism through which QE affects bank lending. That is, capital gains on MMA assets eligible for purchase under the PSPP are significantly associated with increased lending, and undercapitalized banks that held large amounts of such assets increased lending the most. We also test for the salience of the portfolio rebalancing channel and find there is a significant impulse in trading activity after the PSPP announcement among banks with a larger MMA PSPP exposure (i.e., more exposed banks sell more eligible securities), but only in 2019. Even after controlling for this trading activity, we find that lending increased the most among banks that experienced the largest capital gains in

<sup>6</sup> Any demand explanation would thus have to posit that loan demand increased after the PSPP announcement, in both 2015 and 2019. This possibility is highly unlikely and is inconsistent with the reduction in interest rates that we document, as an increase in demand should produce an increase in interest rates.

their MMA portfolio, and the transmission of QE is essentially entirely driven by the recapitalization channel.

Do these results generalize beyond QE? Both QE and other monetary policies, such as forward guidance, influence long-term yields and could affect banks' lending behavior similarly. We thus extend our analysis to study how monetary policy shocks to the long end of the yield curve immediately after the European Central Bank's policy announcements—so-called “*path* rate shocks”—affect bank loan supply between 2010 and 2020. Similar to QE, these more general shocks affect loan supply only through banks' holdings of MMA securities, and the effects are larger for less capitalized banks. For a bank at the median of the MMA exposure, a 1-percentage-point negative shock is associated with a 0.41-percentage-point increase in lending if the bank is at the 25th percentile of the distribution of Tier 1 capital to risk-weighted assets, and 0.29 percentage points if the bank is at the 75th percentile.

These results imply that while HCA can shield bank equity when sovereign assets lose value, HCA also prevents central banks' yield curve management policies from increasing loan supply to firms during periods of distress. We use a simple analytic framework to elaborate this basic intuition and show that a hybrid accounting approach that passes through changes in the risk-free rate, but fixes the sovereign spread at its historical value, can both stabilize banks' equity and allow monetary policy to stimulate loan supply to firms. More generally, our results suggest that HCA can interact with other policies that work through banks' regulatory capital. For instance, we point out that HCA and a static capital requirement—a combination frequently used in practice—is equivalent to MMA combined with a tightening of the capital requirement in response to a reduction in yields.

These results have important implications. The debate over MMA versus HCA in bank regulation has generally focused on the trade-off between the benefits of providing timely information to ensure market discipline versus the costs of excess volatility and its impact on banks' investment decisions and asset prices.<sup>7</sup> Very little attention has been devoted to understanding whether these macroprudential accounting policies influence other macro-stabilization tools, such as central bank yield curve management. To our knowledge, this is the first paper to suggest that attempts to stabilize the banking system through HCA might perversely make monetary policy less effective. Given the increased use of central bank yield curve management policies, the frictions imposed by HCA on bank lending can be sizeable.

A related contribution centers on the efficacy of monetary policy at the zero lower bound. There is considerable debate about the effectiveness of

<sup>7</sup> Some have argued that MMA propagated shocks through the financial system in 2008–2009; see Allen and Carletti (2008) and Plantin, Sapra, and Shin (2008) for models that discuss the potential costs of MMA in bank capital regulation. Ellul et al. (2015) provide evidence on some of the potential drawbacks of HCA in the context of the U.S. insurance industry.

monetary policy when interest rates are close to zero or negative, with many macroeconomic models noting that monetary policy may no longer be effective at the zero lower bound (see the discussion in [McKay, Nakamura, and Steinsson 2016](#)). However, empirical studies, such as [Swanson and Williams \(2014\)](#), suggest that longer-term yields remain sensitive to new information, such as monetary policy announcements, even when the policy rate is at the zero lower bound. Our findings extend those in [Swanson and Williams \(2014\)](#), showing that unconventional monetary policy can directly relieve financing constraints at firms, but only through the effects transmitted by banks' MMA assets.<sup>8</sup>

## 1. Related Literature

A small but growing literature analyzes the effects of macroprudential accounting rules on banks' investment and risk-taking decisions. [Begenau et al. \(2023\)](#) show that accounting rules that allow for delayed loss recognition are crucial to explain banks' response to negative shocks. [Fuster and Vickery \(2018\)](#) and [Kim, Kim, and Ryan \(2019\)](#) find mixed results on the increase in prevalence of MMA on risk-taking decisions, but they document that banks reclassify securities toward sections of their balance sheets that are subject to HCA. [Granja \(2023\)](#) shows that weaker U.S. banks reclassified securities in 2022 to take advantage of HCA. While most of this literature focuses on the effects of macroprudential accounting rules on banks' decisions, our paper is the first—to our knowledge—that studies how these rules shape the effects of other major interventions such as monetary policy and capital requirements.

Our work is related to the literature that studies the interaction between macroprudential and monetary policies. Relative to this literature, our innovation is the focus on accounting-based macroprudential rules, which are widely used in practice. The theoretical arguments in [Di Tella \(2019\)](#) and [Farhi and Werning \(2016\)](#) suggest that macroprudential policies that stabilize banks' net worth can reduce booms and busts, and the joint implementation of monetary and macroprudential policies can further dampen the real effects of adverse shocks (see, for instance, [Van der Groot 2021](#)). [Aiyar, Calomiris, and Wieladek \(2016\)](#) find little evidence of an interaction between conventional monetary policy and capital requirements, and [Altavilla, Laeven, and Peydró \(2020\)](#) and [Bruno, Shim, and Shin \(2017\)](#) study whether accommodative macroprudential environments tend to boost the effects of monetary policy easing. However, they use broad indexes of macroprudential regulation rather

<sup>8</sup> An important literature has examined the impact of monetary policy when rates are negative. For example, [Altavilla et al. \(2022\)](#) find that negative rates can lead to more firm investment. [Bottero et al. \(2022\)](#) show that negative rates increase credit supply through a portfolio rebalancing channel. [Bittner et al. \(2023\)](#) study the interaction between QE and negative rates, noting that negative rates might impair the transmission of QE.

than focusing on the specific channels of the accounting-based macroprudential policies as we do.<sup>9</sup>

We also build on the growing literature on QE. In the case of the PSPP, several studies have focused on the asset price responses to the program's announcement. Notably, the evidence in [Andrade et al. \(2016\)](#) and [Altavilla, Carboni, and Motto \(2021\)](#) suggests that the PSPP's announcement led to a drop in European sovereign yields and an increase in asset prices, including the stock prices of banks. Extrapolating from this asset price response, the model-based results in [Andrade et al. \(2016\)](#) suggest that the PSPP led to an increase in output in part through the bank lending channel. A few papers study important aspects of the PSPP using bank-firm-level data. For instance, [Bittner et al. \(2023\)](#) analyze the interaction with negative interest rates, and [Nogueira, Farinha, and Blattner \(2023\)](#) compare the effects of the PSPP with those of other monetary policies. We focus instead on the interaction with macroprudential accounting regulations. More generally, the transmission of QE onto bank loan supply and the overall efficacy of QE remain open questions; see the survey in [Fabo et al. \(2021\)](#). Our results on the effects of the macroprudential accounting regimes may suggest why the effects of QE can be heterogeneous and difficult to detect.

Beyond micro and macroprudential channels, the evidence in [Peydró, Polo, and Sette \(2021\)](#) observes that the transmission of monetary policy onto bank lending can be impaired if banks hoard liquidity or engage in securities trading instead of lending (see also [Abbassi et al. \[2016\]](#) in the case of German banks). Using Italian micro data similar to ours, [Peydró, Polo, and Sette \(2021\)](#) observe that banks more exposed to the PSPP may have engaged in more securities trading. However, those authors do not directly measure the effects of the PSPP onto loan supply, nor do they study the role of macroprudential accounting regulations.

Our analysis of the bank lending channel of QE is also closely related to work based on U.S. data ([Foley-Fisher, Ramcharan, and Yu 2016](#); [Rodnyansky and Darmouni 2017](#); [Chakraborty, Goldstein, and MacKinlay 2020](#); [Luck and Zimmermann 2020](#)).<sup>10</sup> With detailed supervisory data on banks' asset holdings, we can make progress in understanding the effects of heterogeneity in the application of MMA and HCA in asset valuation. The relatively high frequency of our monthly data also helps us to be clear on the counterfactual and exclude alternative interpretations stemming from other economic news. Moreover, because we observe the near-universe of Italian firms, our results are less likely to be sensitive to the choice of firms in any given sample. More generally, our

<sup>9</sup> Other papers study macroprudential policies alone using credit register data; notably, [Jiménez et al. \(2017\)](#) focus on countercyclical capital buffers. Some macroprudential policies also encompass household leverage (see, e.g., [DeFusco, Johnson, and Mondragon 2020](#)).

<sup>10</sup> A sizeable literature tackles the transmission of QE through the mortgage and housing channel. See, for example, [Di Maggio et al. \(2017\)](#), [Beraja et al. \(2019\)](#), [Di Maggio, Kermani, and Palmer \(2020\)](#), and [Ramcharan \(2020\)](#).



approach builds on the broader literature examining the bank lending channel using micro data, such as [Jiménez et al. \(2014\)](#), and classic studies such as [Kashyap and Stein \(2000\)](#) and [Peek and Rosengren \(2000\)](#).

Finally, because of the link that we establish between the use of HCA to value some assets and capital requirements, our work is also related to some of the existing theoretical and quantitative analysis that links financial regulation with banks' assets holdings. [Van den Heuvel \(2008\)](#) formalizes the idea that higher capital requirements reduce the riskiness of banks' investments, and [Pancost and Robatto \(2023\)](#) quantify this effect to be of first-order importance. [Bahaj and Malherbe \(2020\)](#) show that higher capital requirements can increase bank lending by making banks safer and increasing the loan repayments that benefit shareholders. Our mechanism is complementary in that HCA combined with a static capital requirement is equivalent to MMA and a type of time-varying capital requirement. In this sense, our work is related to the papers that study time-varying capital requirements ([Davydiuk 2017](#); [Malherbe 2020](#); [Canzoneri et al. 2023](#)).

## 2. Background: The Monetary Policy Transmission Mechanism and Macroprudential Accounting Regulations

The broader literature has identified several channels through which various types of QE programs can affect asset prices and bank lending ([Brunnermeier and Sannikov 2013](#); [Krishnamurthy and Vissing-Jorgensen 2011](#)), and this section reviews the main theories. We also summarize basic (macroprudential) accounting rules, which banks are subject to when valuing their security holdings.

A key transmission mechanism is the so-called "recapitalization channel." According to this channel, an increase in security prices triggered by QE would raise the value of banks' securities holdings and banks' net worth. This would allow leveraged-constrained banks that are exposed to the PSPP to expand their loan supply with a stronger effect for undercapitalized banks. This channel is likely to be important in our empirical setting, as the PSPP increased the market value of PSPP-eligible securities ([Andrade et al. 2016](#); [Altavilla, Carboni, and Motto 2021](#)) and Italian banks were undercapitalized in our sample period, especially circa 2015. In our setting, the recapitalization channel predicts that the PSPP's impact on lending should vary with the ex ante cross-sectional differences in banks' holdings of PSPP-eligible securities, as well as with cross-sectional differences in banks' capitalization.

The liquidity channel is another potentially important channel through which the PSPP might affect banks' loan supply. This channel builds on the idea that central bank asset purchases directly increase the liquidity of eligible assets, allowing exposed banks to more easily liquidate such assets to increase lending without generating a price impact. The liquidity channel predicts, in our setting, that the PSPP's impact should vary with the ex ante cross-sectional



differences in banks' liquidity and their PSPP exposure via their holdings of eligible securities.

Another important mechanism is the so-called "portfolio rebalancing channel." According to this channel, banks may sell the appreciated PSPP-eligible securities and rebalance their asset holdings toward investments with a higher expected return, such as loans. In the portfolio rebalancing case, the PSPP's loan supply impact should vary with the cross-sectional differences in ex post asset sales at exposed banks.<sup>11</sup>

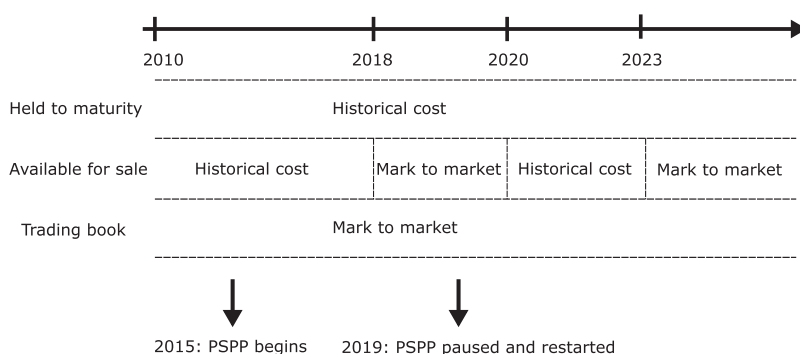
Some channels, such as the recapitalization and portfolio rebalancing channel, might be crucially affected by the pass-through of market prices and yields onto banks' *regulatory* capital. This is the case, for instance, if banks' decisions are affected by regulatory constraints rather than market-based leverage constraints. Because regulatory capital is essentially given by the value of banks' assets net of their liabilities, the prevailing accounting rules used to value banks' assets, including security holdings, can be important for the transmission of monetary policy. We now briefly describe these rules for Italian banks.

A security can be valued using historical cost accounting (HCA) or mark-to-market accounting (MMA) for the purpose of capital requirement regulation. Under HCA, a security's value depends on its purchase price, and changes in market prices do not affect its regulatory net worth. In contrast, under MMA, the security is valued using current market prices, and changes in market conditions pass through onto the bank's regulatory net worth.

Whether a security is valued using MMA or HCA depends on how a bank classifies the security on its balance sheet: holdings of the same securities can be valued differently depending on their classification. In general, banks can classify a security in one of three ways: *held to maturity*, *available for sale*, or *at fair value through profit or loss*. The latter is often referred to as the *trading book*.<sup>12</sup> If the bank plans to hold the security until maturity, it typically classifies the security as held to maturity. If the bank intends to keep the security on its balance sheet but retain the option to sell, the bank classifies the security as available for sale (AFS). Securities that are traded more frequently are classified in the trading book. Key for our analysis is that regulators heavily scrutinize these classifications, so that the classifications are sticky and accurate, as banks cannot easily reclassify securities without

<sup>11</sup> Other channels, such as the signaling channel, are not as easily testable. The signaling channel posits that because QE programs are difficult to reverse, programs like the PSPP signal a central bank's commitment to prolonged low short-term rates. However, the cross-sectional variation in these beliefs is difficult to observe, making it difficult to empirically quantify the importance of channels like signaling.

<sup>12</sup> IFRS 9, which has been applied since 2018, says that an entity can classify its securities in three ways: *amortized cost*, *fair value through other comprehensive income*, and *fair value through profit or loss*. We map these categories onto those of the IAS 39 framework, that is, *held to maturity*, *available for sale*, and *trading book*, respectively.



**Figure 2**  
**Sovereign security classification and prevalent macroprudential accounting regime**

This figure shows how identical sovereign assets on the balance sheet are valued differently depending on their regulatory classification, for the purpose of computing regulatory capital. Between 2010 and 2018, sovereign assets classified as available for sale were valued at historical cost. Between 2018 and 2019, sovereign assets classified as available for sale were then marked to market. In 2020, amid the COVID-19 pandemic, regulators introduced a *prudential filter* that allowed banks to again value available-for-sale sovereign assets at historical cost, based on the market value as of December 31, 2019. Assets classified as held to maturity are always valued at historical cost, whereas those in the trading book are always marked to market.

prior regulatory approval.<sup>13</sup> In our empirical setting, we show that holdings of PSPP-eligible securities are indeed very sticky over time, and we conduct tests to rule out the possibility that banks select into exposure in anticipation of the PSPP announcement (see Section 4.3.2).

Figure 2 summarizes the macroprudential accounting rules for Italian banks. The rules governing sovereign held-to-maturity and trading book securities have been relatively unchanged. Held-to-maturity securities are valued using HCA, and trading book securities are valued using MMA. But for AFS securities issued by euro area central governments—which are eligible for purchase under the PSPP—regulators have applied different regimes over time. To insulate the banking system from the effects of the sovereign crisis, during the early 2010s and until the end of 2017, regulators permitted banks to use HCA to value AFS sovereign securities.<sup>14</sup> But as of 2018, regulators required AFS securities to be valued using MMA.<sup>15</sup>

<sup>13</sup> Under the IAS 39 rules in place until 2018, the reclassification of a held-to-maturity security is generally not allowed, and if it takes place, it can trigger the reevaluation of all banks' holdings based on market prices. This trigger is part of the so-called "tainting rule." Under IFRS 9, in place since 2018, reclassifications are only possible when an entity changes its business model for managing financial assets, such as in the event of mergers and acquisitions. In the case of available for sale versus trading book, if a bank were to classify securities as available for sale but then trade them frequently—as if they were in the trading book—regulators can flag this behavior and take corrective actions.

<sup>14</sup> Formally, AFS securities were always evaluated using MMA, but between 2010 and 2017, regulators allowed banks to omit unrealized gains and losses from the computation of regulatory capital, which is equivalent to using HCA for the purpose of computing regulatory capital. This is analogous to the so-called "AOCI filter" in the United States. See CEBS (2004), Regulation EU no. 575/2013 (page 508), Article 467, and Bank of Italy (2010).

<sup>15</sup> As the COVID-19 crisis hit in 2020—after the end of our sample period—European regulators allowed banks to again use HCA for available-for-sale securities; see Bank of Italy (2020).

This regulatory structure creates variation in the accounting rules used to classify sovereign securities, both at any given time within the cross-section of securities and over time for the AFS classification. Our analysis exploits these sources of variation to calculate a bank's exposure to the PSPP and other central bank shocks to the yield curve in order to understand how these policies affect bank credit supply. The first PSPP announcement took place in January 2015, when only trading book sovereign securities were valued at MMA. But the 2019 resumption of the PSPP occurred when both AFS and trading book sovereign securities were valued at MMA. Thus, in addition to the fact that banks had different holdings of HCA and MMA securities at the time of each announcement, we also exploit the change in the classification of AFS sovereign assets from HCA to MMA between 2015 and 2019 to identify the role of macroprudential accounting rules in mediating the effects of the PSPP.

### 3. Data

On January 22, 2015, the European Central Bank (ECB) announced the PSPP program—to be started in March—consisting of the purchases of about €50 billion per month of euro-denominated debt securities issued by euro area central governments and supranational institutions, with residual maturity between 2 and 30 years.<sup>16</sup> The program was paused at the end of December 2018, and the ECB announced on September 12, 2019, a restart as of November of that year.<sup>17</sup> This subsection describes the data we use to measure banks' exposure to the PSPP and identify the transmission mechanism. We draw on several sources, primarily collected by the Bank of Italy.

First, we use the Italian Central Credit Register, which contains information for each firm-bank lending relationship. The credit register is a monthly panel data set at the firm-bank-time level. That is, each entry denotes the amount of lending made by a particular bank to a particular nonfinancial firm in a given month. All loans above €30,000 are included in the register, thereby making the coverage near-universal. Second, we use AnaCredit and Taxia, two data sets with information collected by the Bank of Italy, to retrieve interest rate data on new term loans. AnaCredit has data at a monthly frequency but only covers 2019, whereas Taxia has data for 2015 but only at a quarterly frequency. Third, we use the Bank of Italy Credit and Financial Institutions' Supervisory Reports to obtain banks' balance sheet data. A key element of these data is represented by banks' security holdings at the ISIN level, and for each ISIN,

<sup>16</sup> Because the exact criteria for international institutions were not fully clarified at the time, we consider as eligible the debt of supranational institutions with main headquarters in a euro area country. See the technical annex from the European Central Bank's (2015) press release. In addition, the securities had to fulfil the collateral eligibility criteria for the ECB and must have had a credit rating of at least CQS3 (i.e., Credit Quality Step 3). This corresponds to at least BBB- for S&P's and Fitch and Baa3 for Moody's.

<sup>17</sup> In 2019, purchases were estimated at €20 billion per month, and debt issued by euro area local governments was eligible to be purchased under the program, too. See Decision (EU) 2015/2464 of the European Central Bank.

the breakdown of the holdings classified as held to maturity, available for sale, or in the trading book. This level of granularity allows us to construct each bank's measure of exposure to the 2019 and 2015 QE announcements and, together with security prices from the Centralised Securities Database of the Eurosystem, banks' trading activity.

In most of our analyses, we work with monthly data over a 12-month window around the QE announcement dates (i.e., around September 12, 2019, and January 22, 2015). We focus on banking groups (hereinafter referred to as "banks") for which the bank holding company is a joint stock company. That is, we exclude mutual and cooperative banks because they are subject to different regulations. We also drop foreign banks, leaving our final sample at 90 banks for 2019 and 95 banks for 2015. Our focus on banking groups (as opposed to single banks) is motivated by the fact that regulators set key regulations, such as capital requirements, at the group level.<sup>18</sup> Our approach is similar to that used in recent papers with Italian banking data, such as [Benetton and Fantino \(2021\)](#) and [Bottero et al. \(2022\)](#).

Table 1 provides key summary statistics about the firm-bank lending relationships and the interest rate data. Our main analysis uses the monthly growth rate of the amount borrowed for each firm-bank pair with an ongoing lending relationship. Focusing on firms with at least two lending relationships ([Khwaja and Mian 2008](#)), we have access to more than 8 million observations for the 2019 episode and about 6 million for 2015.

Table 2 contains summary statistics about the banks in our sample. Holdings of PSPP-eligible securities were substantial. Banks held on average 17.06% and 11.44% of assets in securities eligible to be purchased under the PSPP prior to the 2019 and 2015 announcements, respectively. Holdings of eligible securities valued at historical cost were 11.12% and 11.00% in 2019 and 2015, respectively, and holdings of eligible MMA securities were 6.00% and 0.45% in 2019 and 2015, respectively. The 2015 mark-to-market figure appears small, as it includes only securities in the trading book as discussed in Section 2, but there is significant heterogeneity across banks that gives rise to a large cross-sectional variation: the coefficient of variation is in fact higher in 2015 than in 2019.

#### 4. Macroprudential Regime and the Impact of QE on Credit Supply

This section shows that the macroprudential accounting regime mediates the impact of the PSPP on bank lending. We begin with the 2019 PSPP announcement, when mark-to-market accounting (MMA) applied to

<sup>18</sup> Because of the regulatory approach, banks within the group do not need to meet capital requirements individually. This implies, for instance, that a credit expansion can be carried out only by some banks that are part of the group, and any within-group borrowing and lending would cancel out when regulators check group-level capital ratios.

**Table 1**  
**Summary statistics, firm-bank lending relationships**

*A. Number of lending relationships*

|       | August 2019 |            | December 2014 |            |
|-------|-------------|------------|---------------|------------|
|       | N           | Percentage | N             | Percentage |
| 2     | 177,719     | 60         | 141,323       | 65         |
| 3     | 62,810      | 21         | 44,983        | 21         |
| 4     | 26,928      | 9          | 17,089        | 8          |
| 5     | 13,101      | 4          | 7,330         | 3          |
| 6+    | 17,331      | 6          | 5,993         | 3          |
| Total | 297,889     | 100        | 216,718       | 100        |

*B. Firm-bank lending relationships*

| Firm-bank lending relationships                                            | N          | Mean    | 1st quartile | Median  | 3rd quartile | Std. dev. |
|----------------------------------------------------------------------------|------------|---------|--------------|---------|--------------|-----------|
| <i>All firms, 2019 PSPP</i>                                                |            |         |              |         |              |           |
| Amount borrowed, EUR (as of August 2019)                                   | 1,628,965  | 368,983 | 32,474       | 71,806  | 192,770      | 5,241,722 |
| log change amount borrowed, % (12-month window)                            | 17,130,567 | -0.56   | -3.33        | -0.33   | 0.38         | 50.56     |
| <i>Firms with more than one bank relationship in each month, 2019 PSPP</i> |            |         |              |         |              |           |
| Amount borrowed, EUR (as of August 2019)                                   | 846,286    | 533,090 | 33,061       | 93,089  | 287,670      | 6,942,719 |
| log change amount borrowed, % (12-month window)                            | 8,780,566  | -0.77   | -5.72        | -0.32   | 1.99         | 57.20     |
| <i>All firms, 2015 PSPP</i>                                                |            |         |              |         |              |           |
| Amount borrowed, EUR (as of December 2014)                                 | 1,216,779  | 414,973 | 36,370       | 79,218  | 211,910      | 5,444,324 |
| log change amount borrowed, % (12-month window)                            | 13,186,871 | 0.17    | -2.04        | 0.00    | 0.61         | 46.33     |
| <i>Firms with more than one bank relationship in each month, 2015 PSPP</i> |            |         |              |         |              |           |
| Amount borrowed, EUR (as of December 2014)                                 | 563,922    | 625,108 | 43,978       | 115,779 | 346,191      | 7,587,313 |
| log change amount borrowed, % (12-month window)                            | 6,117,128  | 0.19    | -2.97        | 0.00    | 1.24         | 51.16     |

*C. Interest rates on new term loans*

| Interest rates on new term loans                                                | N       | Mean | 1st quartile | Median | 3rd quartile | Std. dev. |
|---------------------------------------------------------------------------------|---------|------|--------------|--------|--------------|-----------|
| <i>All observations, 2019 PSPP</i>                                              |         |      |              |        |              |           |
| Interest rate, % (12-month window)                                              | 248,439 | 4.18 | 2.44         | 3.80   | 5.52         | 2.35      |
| <i>Firms with new term loans from multiple banks in each month, 2019 PSPP</i>   |         |      |              |        |              |           |
| Interest rate, % (12-month window)                                              | 14,580  | 3.17 | 1.63         | 2.67   | 4.18         | 2.10      |
| <i>All observations, 2015 PSPP</i>                                              |         |      |              |        |              |           |
| Interest rate, % (12-month window)                                              | 127,369 | 4.82 | 3.07         | 4.35   | 6.13         | 2.46      |
| <i>Firms with new term loans from multiple banks in each quarter, 2015 PSPP</i> |         |      |              |        |              |           |
| Interest rate, % (12-month window)                                              | 9,451   | 3.97 | 2.39         | 3.54   | 5.03         | 2.19      |

Panel A displays the distribution of the number of lending relationships for the firms with outstanding loans reported in the Italian credit register. Panel B displays the distribution of the amount borrowed and the log change in the amount borrowed. Panel C displays the interest rate distribution for new term loans.

PSPP-eligible securities in the available-for-sale (AFS) and trading book portfolios. We then move to the 2015 PSPP announcement, when regulators valued sovereign AFS securities using historical cost accounting (HCA) and, among PSPP-eligible, only those in the trading book were valued at MMA. The section concludes with a formal test of the importance of MMA in the transmission mechanism across the two time periods and a battery of robustness tests.

**Table 2**  
**Bank-level summary statistics**

|                                                                       |                      | Mean   | 1st quartile | Median | 3rd quartile | Std dev | Coeff var |
|-----------------------------------------------------------------------|----------------------|--------|--------------|--------|--------------|---------|-----------|
| Eligible securities, % of assets                                      | Aug 2019             | 17.06  | 7.14         | 15.14  | 24.42        | 13.39   | 0.79      |
|                                                                       | Dec 2014             | 11.44  | 1.69         | 8.10   | 18.37        | 11.35   | 0.99      |
|                                                                       | Aug 2019             | 6.00   | 0.18         | 3.16   | 6.66         | 9.74    | 1.62      |
|                                                                       | Dec 2014             | 0.45   | 0.00         | 0.00   | 0.001        | 1.63    | 3.65      |
| Eligible securities, historical cost, % assets                        | Aug 2019             | 11.12  | 3.09         | 7.68   | 18.18        | 10.41   | 0.94      |
|                                                                       | Dec 2014             | 11.00  | 1.33         | 8.10   | 17.46        | 11.18   | 1.02      |
|                                                                       | Aug 2019             | 6.00   | 0.18         | 3.16   | 6.66         | 9.74    | 1.62      |
|                                                                       | Dec 2014             | 10.42  | 0.17         | 6.69   | 17.65        | 11.02   | 1.06      |
| Eligible securities, available for sale and trading book, % of assets | Aug 2019             | 3.65   | 0.29         | 1.36   | 4.69         | 5.57    | 1.53      |
|                                                                       | Dec 2014             | 9.01   | 1.47         | 5.66   | 12.68        | 10.64   | 1.18      |
|                                                                       | Aug 2019             | 4.81   | 0.15         | 2.24   | 6.59         | 7.11    | 1.48      |
|                                                                       | Dec 2014             | 4.12   | 0.00         | 0.19   | 6.30         | 6.46    | 1.57      |
| Noneligible securities, trading book, % of assets                     | Aug 2019             | 21.85  | 20.25        | 21.53  | 23.39        | 2.16    | 0.10      |
|                                                                       | Dec 2014             | 21.38  | 19.41        | 21.45  | 22.82        | 2.20    | 0.10      |
|                                                                       | Aug 2019             | 25.57  | 7.88         | 22.72  | 32.11        | 38.59   | 1.51      |
|                                                                       | Dec 2014             | 25.24  | 10.85        | 26.75  | 37.71        | 15.59   | 0.62      |
| Business loans, % of assets                                           | Aug 2019             | 3.25   | 0.27         | 1.19   | 3.11         | 6.67    | 2.05      |
|                                                                       | Dec 2014             | 1.31   | 0.08         | 0.76   | 1.29         | 1.82    | 1.39      |
|                                                                       | Jun 2019             | 19.14  | 12.74        | 15.17  | 19.05        | 12.58   | 0.66      |
|                                                                       | Dec 2014             | 22.10  | 10.90        | 14.18  | 26.00        | 24.56   | 1.11      |
| Tier 1 capital, % of risk-weighted assets                             | Feb 2019             | 9.52   | 5.83         | 9.98   | 12.60        | 5.17    | 0.54      |
|                                                                       | Dec 2014             | 0.33   | 0.00         | 0.00   | 1.00         | 2.20    | 6.65      |
|                                                                       | Jun 2019             | 4.08   | -0.64        | 0.57   | 4.29         | 11.23   | 2.75      |
|                                                                       | Mar 2014             | -0.35  | -3.73        | 0.00   | 2.97         | 12.32   | -35.64    |
| TLTRO exposure, % of total assets                                     | Aug 2019             | 0.48   | 0.00         | 0.00   | 1.00         | 0.50    | 1.05      |
|                                                                       | Aug 2014             | 0.28   | 0.00         | 0.00   | 1.00         | 0.45    | 1.60      |
|                                                                       | Aug 2019             | 1.64   | 0.00         | 0.00   | 1.33         | 3.58    | 2.19      |
|                                                                       | Aug 2014             | 1.68   | 0.00         | 0.00   | 1.53         | 3.51    | 2.09      |
| Net interbank position, % of assets                                   | Aug 2019             | 0.09   | 0.00         | 0.00   | 0.06         | 0.25    | 2.65      |
|                                                                       | Jun to Sept 2019     | 0.80   | 0.00         | 0.00   | 0.91         | 1.68    | 2.12      |
|                                                                       | Mar 2019 to Feb 2020 | -2.64  | 0.00         | 0.00   | 0.00         | 183.64  | -69.54    |
|                                                                       | Jul 2014 to Jun 2015 | -12.11 | 0.00         | 0.00   | 0.00         | 268.87  | -22.21    |

The table displays key summary statistics for the banks in our sample. We have 95 banks in 2014–2015 and 90 in 2019–2020. Tier 1 data are available for 89 banks in 2019–2020 and 93 banks in 2014–2015. Cash is from end-of-year balance sheets, and thus, measured at year-end prior to each announcement. See [Internet Appendix II](#) for the definition of TLTRO exposure, net interbank position, securitization dummy, and unused allowance.

#### 4.1 The effects of the PSPP: The basic results

This section presents the basic relationship between the PSPP and bank lending, beginning with the 2019 announcement and then repeating the analysis for the 2015 announcement. Informed by theories of the monetary transmission mechanism, the research design combines the cross-sectional variation in banks' exposure to the PSPP via their holdings of MMA-eligible securities with monthly data on lending. The research design then uses a difference-in-difference framework to exploit the discrete nature of the ECB's PSPP announcements. This difference-in-difference framework employs a narrow time window around each PSPP announcement with monthly data to isolate the effects of the PSPP from other ongoing ECB policies, such as forward guidance, negative interest rates, and the targeted longer-term refinancing operations.<sup>19</sup> For each announcement, the baseline estimation window spans both the 6 months before and 6 months after the announcement.

The design also uses firm-by-year-month fixed effects to absorb nonparametrically loan demand at the firm-by-year-month to identify the PSPP-induced bank lending supply response (Khwaja and Mian 2008). Additionally, because the Italian credit register contains the near-universe of the bank-firm credit relationships, we can measure the effects of QE on business lending more completely than inferring a treatment effect based on a selected sample of larger firms, usually from DealScan or through regulatory data in the United States.

The dependent variable is the monthly growth rate in lending at the intensive margin, namely, the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  between month  $t$  and  $t-1$ ,  $\log(L_{b,f,t}) - \log(L_{b,f,t-1})$ . The baseline specification for the 2019 announcement is

$$\begin{aligned} \Delta \log L_{b,f,t} = & \sum_{\tau \neq 2019m8} \beta_{\tau} \times I_{\tau} \times QE_b \\ & + \sum_{\tau \neq 2019m8} \gamma_{\tau} \times I_{\tau} \times Y_b + \delta Z_{b,t} + \psi_b + \psi_{f,t} + \varepsilon_{b,f,t}. \end{aligned} \quad (1)$$

The variable  $QE_b$  is a measure of a bank's exposure to the PSPP based on the bank's holdings of securities eligible for purchase under the PSPP. We compute  $QE_b$  in the month before the announcement, so in the case of the September 2019 announcement, we use banks' asset holdings at the end of August 2019. We then interact  $QE_b$  with a set of time dummies  $I_{\tau}$ , one for each month, dropping the one that corresponds to the preannouncement month. Thus, our coefficients of interest are the  $\beta_{\tau}$ . This sequence captures the effects of the PSPP announcement on bank lending supply and allows us to check for

<sup>19</sup> See, for example, McKay, Nakamura, and Steinsson (2016), Andreeva and García-Posada (2021), Bottero et al. (2022), and Benetton and Fantino (2021).



any possible pre-trend in lending growth. The terms  $\psi_b$  and  $\psi_{f,t}$  are bank and firm-by-time fixed effects, respectively, and standard errors are clustered at the bank level.

To exclude alternative interpretations, the baseline specification includes three controls, captured by the vector  $Y_b$  and the term  $Z_{b,t}$  in Equation (1). First, we include bank size using the logarithm of total assets as of the month before the announcement, interacted with time dummies centered on the announcement month. Second, because banks' holdings of central bank reserves as well as borrowing from the ECB have been shown to independently affect bank behavior (Diamond, Jiang, and Ma 2024; Drechsler et al. 2016), we include central bank reserves (defined as cash plus deposits at the central bank as a fraction of total assets, computed before the announcement and interacted with time dummies centered on the announcement month) and the logarithmic change in the euro value that bank  $b$  borrows from the ECB at time  $t$ , denoted by  $Z_{b,t}$ . Additional controls, including those related to bank characteristics and banks' exposure to other ECB policies, are added in the robustness checks of Section 4.3.

We first follow the literature and define  $QE_b$  using the broadest exposure definition: the ratio of a bank's securities that are eligible to be purchased under the PSPP to total bank assets (Rodnyansky and Darmouni 2017; Luck and Zimmermann 2020). This broad exposure measure ignores the macroprudential-based accounting treatment of sovereign assets. Italian banks hold large amounts of sovereign securities, and in the month before the 2019 announcement, they had on average 17.1% of assets in PSPP-eligible securities.

Table 3 presents the results. We begin by showing all the event-study coefficients in Equation (1) to be transparent, before moving to a more concise difference-in-difference specification for ease of exposition. From column 1 of Table 3, there is no evidence that broad exposure to the PSPP is associated with an increase in lending after the announcement. The point estimates are small and show a pre-trend, and there is no clear difference between the pre- and post-announcement periods.<sup>20</sup>

Next, we redefine the measure of exposure  $QE_b$  to consider the accounting-based macroprudential regulation in place in 2019. Specifically, we include only eligible securities that are valued at MMA. Based on the 2019 macroprudential rules, sovereign securities were valued at MMA if classified as either AFS or held in the trading book, which jointly account for 6.00% of total assets on average across banks.

The results of using only MMA eligible securities—shown in column 2—dramatically differ from those in column 1. Column 2 of Table 3 shows that

<sup>20</sup> There is a significant pre-trend (average March to July 2019: 0.06,  $p$ -value = .01), and the coefficient for the announcement month of September 2019 is not statistically different from the March to July 2019 average (difference = 0.05,  $p$ -value = .10).

Table 3  
The impact of the PSPP announcement on bank lending

|                                                    | September 2019 announcement |                            |                            |                    | January 2015 announcement |                            |                            |                   |
|----------------------------------------------------|-----------------------------|----------------------------|----------------------------|--------------------|---------------------------|----------------------------|----------------------------|-------------------|
|                                                    | (1)                         | (2)                        | (3)                        | (4)                | (5)                       | (6)                        | (7)                        | (8)               |
| [Month -6] $\times QE_b$                           | -0.008<br>[0.015]           | 0.021<br>[0.078]           | 0.845<br>[0.970]           |                    | -0.050<br>[0.040]         | 0.079<br>[0.133]           | 0.362<br>[0.633]           |                   |
| [Month -5] $\times QE_b$                           | 0.147***<br>[0.036]         | 0.106<br>[0.161]           | -0.197<br>[1.739]          |                    | -0.156***<br>[0.056]      | 0.183<br>[0.180]           | 0.439<br>[0.635]           |                   |
| [Month -4] $\times QE_b$                           | 0.054**<br>[0.025]          | 0.084<br>[0.068]           | 0.833<br>[0.689]           |                    | -0.003<br>[0.055]         | 0.278<br>[0.234]           | 0.836<br>[0.582]           |                   |
| [Month -3] $\times QE_b$                           | 0.004<br>[0.024]            | 0.042<br>[0.125]           | 1.349<br>[1.554]           |                    | -0.089<br>[0.055]         | -0.105<br>[0.243]          | -0.910<br>[0.938]          |                   |
| [Month -2] $\times QE_b$                           | 0.115***<br>[0.033]         | 0.045<br>[0.149]           | 0.028<br>[1.715]           |                    | -0.010<br>[0.063]         | 0.232<br>[0.284]           | -0.363<br>[0.925]          |                   |
| <b>Announcement month <math>\times QE_b</math></b> | <b>0.111**<br/>[0.047]</b>  | <b>0.244**<br/>[0.116]</b> | <b>2.610**<br/>[1.038]</b> |                    | <b>-0.080<br/>[0.054]</b> | <b>0.437**<br/>[0.182]</b> | <b>1.759**<br/>[0.733]</b> |                   |
| [Month +1] $\times QE_b$                           | 0.127**<br>[0.061]          | -0.071<br>[0.152]          | 1.136<br>[1.501]           |                    | -0.078**<br>[0.039]       | 0.139<br>[0.187]           | 0.022<br>[0.703]           |                   |
| [Month +2] $\times QE_b$                           | -0.016<br>[0.029]           | 0.111<br>[0.114]           | 1.928<br>[1.501]           |                    | 0.009<br>[0.055]          | 0.058<br>[0.171]           | 0.368<br>[0.558]           |                   |
| [Month +3] $\times QE_b$                           | 0.104**<br>[0.040]          | 0.339***<br>[0.107]        | 3.367***<br>[0.700]        |                    | -0.085*<br>[0.045]        | 0.084<br>[0.172]           | 0.022<br>[0.569]           |                   |
| [Month +4] $\times QE_b$                           | 0.111***<br>[0.035]         | 0.071<br>[0.146]           | -0.455<br>[1.758]          |                    | -0.119**<br>[0.051]       | 0.039<br>[0.194]           | -0.563<br>[0.766]          |                   |
| [Month +5] $\times QE_b$                           | -0.024<br>[0.015]           | 0.031<br>[0.051]           | -0.161<br>[0.379]          |                    | 0.007<br>[0.055]          | 0.055<br>[0.138]           | 0.207<br>[0.504]           |                   |
| [Post] $\times QE_b$                               |                             |                            |                            | 0.074**<br>[0.037] |                           |                            |                            | 0.080*<br>[0.048] |
| Eligible securities in $QE_b$                      | All                         | MMA only                   | MMA only (dummy)           | MMA only           | All                       | MMA only                   | MMA only (dummy)           | MMA only          |
| Bank FEs                                           | Yes                         | Yes                        | Yes                        | Yes                | Yes                       | Yes                        | Yes                        | Yes               |
| Firm-time FEs                                      | Yes                         | Yes                        | Yes                        | Yes                | Yes                       | Yes                        | Yes                        | Yes               |
| Size                                               | Yes                         | Yes                        | Yes                        | Yes                | Yes                       | Yes                        | Yes                        | Yes               |
| ECB lending                                        | Yes                         | Yes                        | Yes                        | Yes                | Yes                       | Yes                        | Yes                        | Yes               |
| Reserves                                           | Yes                         | Yes                        | Yes                        | Yes                | Yes                       | Yes                        | Yes                        | Yes               |
| Observations                                       | 8,346,925                   | 8,346,925                  | 8,346,925                  | 8,346,925          | 5,867,308                 | 5,867,308                  | 5,867,308                  | 5,670,501         |
| R-squared                                          | .370                        | .370                       | .370                       | .369               | .394                      | .394                       | .394                       | .395              |

The dependent variable is the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  in month  $t$  (in percentages). Columns 1–4 refer to the 2019 PSPP announcement, and columns 5–8 to the 2015 announcement. In columns 1 and 5 the QE exposure variable,  $QE_b$ , does not distinguish between MMA and HCA accounting. Columns 2 and 6 measure  $QE_b$  using only MMA assets. Columns 3 and 7 use a dummy measure of MMA exposure. Column 3 compares banks in the top tercile of the MMA exposure distribution with those in the bottom tercile, and column 7 compares banks in the top 15% with those in the bottom 85%. Columns 4 and 8 use a “post” indicator that equals 1 in the announcement month and the following months. In column 8, the sample is restricted to bank-firm relationships present in December 2014 and January 2015. Standard errors are clustered at the bank level. Boldface corresponds to the month in which the policy is announced. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

banks exposed to the PSPP via their holdings of MMA securities increased lending almost immediately upon the PSPP’s announcement. This measure of exposure also shows no trend difference in lending growth between exposed and nonexposed banks in the period before the PSPP announcement (average March to July 2019 = 0.04,  $p$ -value = .48). The September 2019 coefficient is also significantly greater than the March–July 2019 average (difference: 0.18,  $p$ -value < .01). Column 2 also shows a positive effect in December 2019

(i.e., 1 month after the actual restart of the purchases) and no subsequent reversal in loan growth.<sup>21</sup>

A back-of-the-envelope calculation using the estimates in column 2 suggests that the PSPP increased loan supply by about €12 billion in this sample period. We obtain this number based on the counterfactual in which all the banks have zero exposure, and conservatively use the lending responses in September and December 2019, that is, the only 2 months with a statistically significant coefficient. The effect is slightly higher at €13 billion if we instead use the average lending response between September and December 2019 (average September to December 2019 = 0.156,  $p$ -value = .048).<sup>22</sup> A similar calculation implies that if all banks' holdings of eligible securities had been valued according to MMA, the overall effect on lending would have been almost three times bigger at €32.6 billion.

Column 3 addresses possible measurement errors in the PSPP exposure. It uses a dummy variable, comparing banks in the top tercile of the exposure distribution with those in the bottom tercile, as in, for example, Chakraborty, Goldstein, and MacKinlay (2020). The results are unchanged. The most exposed banks—those in the top tercile—increased loan supply by 2.61 percentage points right after the 2019 PSPP announcement and by 3.37 percentage points in December 2019, relative to less exposed banks. We again find no significant difference in loan growth in the period before the announcement: the March to August 2019 average is 0.57 ( $p$ -value = .42).

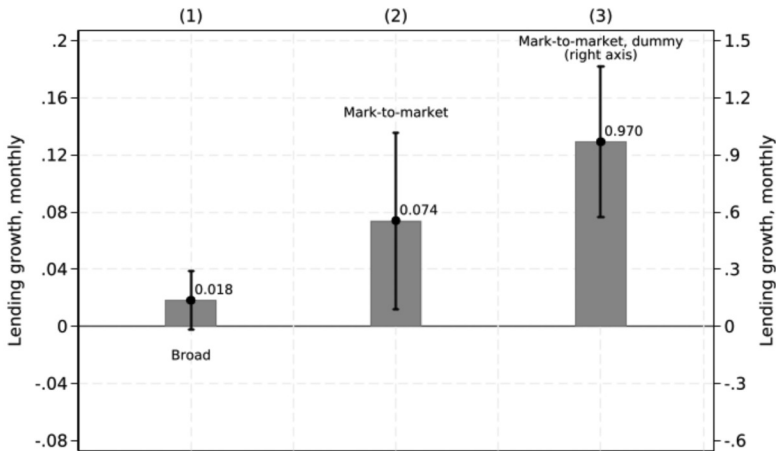
As a further check, column 4 uses a more concise difference-in-difference approach. This is a simple *post* indicator variable interacted with the MMA exposure to compute the average impact of the 2019 PSPP on bank lending over the 6 months after the announcement (i.e., September 2019 to February 2020). Column 4 shows that the interaction term is significant, suggesting that while the lending response was immediate, it also appears persistent over the 6 months. Panel A of Figure 3 provides a simple visualization of the results of the difference-in-difference approach that replicates columns 1–3 of Table 3. The results are essentially unchanged.

The remaining columns of Table 3 repeat these exercises for the 2015 PSPP announcement, using a sample centered around the January 2015 announcement. Unlike 2019, in 2015 regulators used MMA only for trading book assets; sovereign assets classified as AFS—most sovereign assets—were

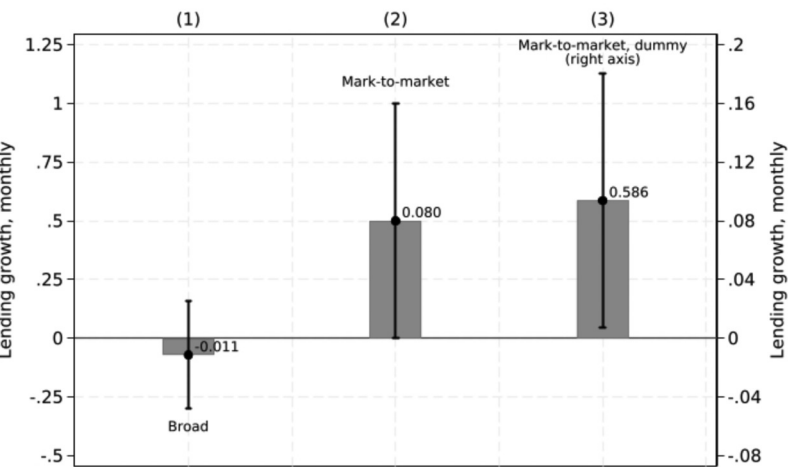
<sup>21</sup> In Internet Appendix I, we provide an additional test that includes both the HCA and MMA exposure measures before the 2019 announcement in a single specification. The coefficients for the MMA exposure variables are unchanged from those in column 2 of Table 3. The post-announcement coefficients of the HCA exposure are not statistically different from those in the prior months. And the difference between the MMA and HCA coefficients is 0.202 for September 2019 ( $p$ -value = .04) and 0.325 for December 2019 ( $p$ -value < .01). We also consider a test that includes both the MMA exposure (column 2) and the broad exposure (column 1) in 2019. The impact of MMA exposure remains similar, but these estimates are noisier because of multicollinearity, as the broad exposure mechanically includes MMA exposure. See Internet Appendix I for additional details and discussions.

<sup>22</sup> In Internet Appendix V, we show that the increase in lending by more exposed banks did not crowd out lending by less exposed ones, so the €12 billion figure represents a net increase in loan supply.

**A 2019 announcement**



**B 2015 announcement**



**Figure 3**  
**The impact of the PSPP announcements, *post* indicator**

Panel A replicates columns 1–3 of Table 3, and panel B replicates columns 5–7 of Table 3, using a *post* indicator variable interacted with the exposure measures. The figure plots the point estimates and 90% confidence intervals. The specification in (1) uses the broad measure of exposure (i.e., all eligible securities) as a fraction of total assets (in percentages), (2) uses the MMA exposure as a fraction of total assets (in percentages), and (3) uses the MMA exposure dummy described in Section 4.1. All the regressions include controls for size, ECB lending, and central bank reserves, as well as bank and firm-time fixed effects. In panel B, the sample is restricted to bank-firm relationships present in December 2014 and January 2015. Standard errors are clustered at the bank level.

valued at HCA in 2015. This difference means that the set of banks that were highly exposed to the 2019 PSPP via their MMA holdings of eligible securities differ from the highly exposed set in 2015. This variation helps address identification concerns based on unobserved bank business model characteristics that might correlate with both the MMA exposure and the lending of a bank, helping us triangulate the role of the macroprudential regime itself in transmitting yield curve shocks onto bank lending.

Focusing on the 2015 PSPP announcement, column 5 of Table 3 replicates the specification in column 1. As in 2019, broad exposure to the PSPP is not associated with an increase in loan supply after the 2015 announcement.<sup>23</sup> Column 6 of Table 3 uses banks' exposure to the PSPP based only on MMA eligible securities according to the 2015 macroprudential accounting framework (i.e., those classified in the trading book). Banks more exposed to the PSPP via MMA securities increased lending immediately upon the PSPP announcement. There is again no pre-trend (average July to November 2014: 0.13,  $p$ -value = .50), and the January 2015 coefficient is significantly greater than the July to November 2014 average (difference: 0.30,  $p$ -value < .01).<sup>24</sup>

Column 7 of Table 3 uses an exposure dummy to deal with possible measurement error. In 2015, holdings of MMA eligible securities were limited to the trading book, and thus, much smaller than in 2019 and with a highly skewed distribution. Hence, we define the 2015 exposure dummy as equal to one if a bank is in the top 15% of the distribution; this corresponds approximately to the median of the distribution of MMA eligible securities relative to total assets, conditional on strictly positive holdings. The results remain robust to using the dichotomous measure of exposure. Loan growth increases by about 1.76 percentage points for highly exposed banks in January 2015, relative to less exposed ones.<sup>25</sup>

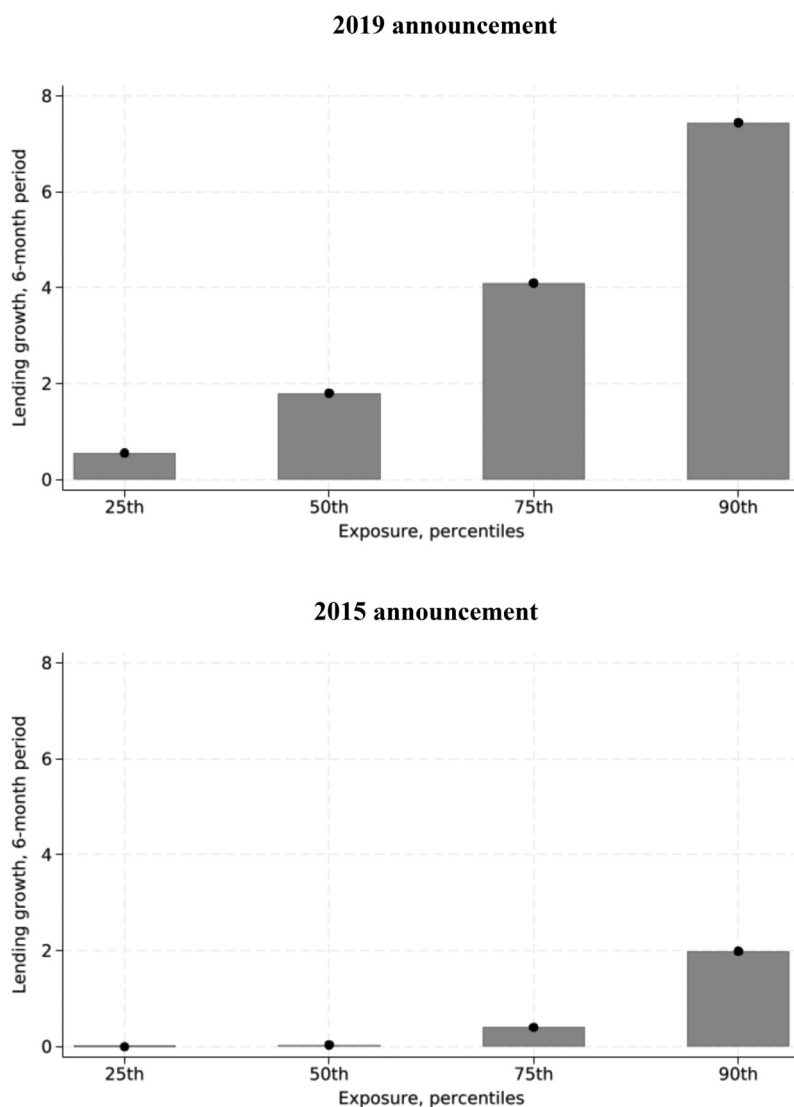
Finally, column 8 of Table 3 uses the more concise difference-in-difference specification, interacting the MMA exposure measure with a *post* indicator variable. The interaction is significant, suggesting that the lending impact lasted over the estimation window, even for 2015. And panel B of Figure 3 visualizes the results of replicating columns 5–7 with the *post* indicator. The results are again essentially unchanged.

Despite the similarity of the 2015 results with those of the 2019 analysis, the implied effect of the 2015 PSPP announcement on loan supply is much smaller. The coefficients in column 6 of Table 3 imply an increase in lending

<sup>23</sup> We calculate the exposure measures as of December 2014 (i.e., the month before the announcement), in line with the approach used for the 2019 PSPP announcement.

<sup>24</sup> Internet Appendix I shows that when directly including both HCA and MMA exposure measures in 2015, the coefficients for the MMA exposure variables are unchanged from those in column 6 of Table 3, while the HCA coefficients are insignificant. Similarly, the MMA coefficients do not change when including both broad and MMA exposure measures.

<sup>25</sup> There is again no significant difference in loan growth across banks in the period before the announcement: the average of the PSPP exposure coefficients in July to November 2014 is 0.07 ( $p$ -value = .90).



**Figure 4**

**Effect of the PSPP announcement on bank loan supply**

The figure plots the impact of the PSPP on bank lending over the 6-month period after the announcement (i.e., September 2019 to February 2020 for the 2019 announcement, and January to June 2015 for the 2015 announcement), as a function of a bank's mark-to-market exposure for banks with a positive exposure.

of only €214 million in 2015, in contrast to the €12 billion in 2019 from column 2. The difference between 2019 and 2015 is large not just for the Italian banking system as a whole but also at every point in the distribution of exposure: compare panels A and B of Figure 4, which plot the effect of

the PSPP in the cross-section of banks with positive exposure. The very small 2015 lending effect reflects the narrow application of MMA in that period's macroprudential accounting regime. That is, sovereign securities were valued at MMA in 2015 only if held in the trading book. In contrast, the 2019 macroprudential regulation required banks to use MMA for both AFS and trading book sovereign securities. Because banks record a significant amount of sovereign assets as AFS, the 2019 PSPP announcement induced a much bigger lending response relative to 2015.

In sum, the evidence suggests that the PSPP elicited a stronger lending response when a larger fraction of a bank's eligible securities was valued at MMA. Using the fact that the only difference between the 2015 and 2019 regimes is the treatment of AFS sovereign assets, the next subsection nests the 2019 and 2015 specifications to formally test the importance of MMA in the transmission of the PSPP.

#### 4.2 Combining the 2019 and 2015 samples: A formal test of the role of the macroprudential accounting regime

This subsection combines the 2015 and 2019 samples to formally test the role played by MMA in the transmission of the PSPP. This combination results in a single panel, where each subpanel spans a 12-month window centered on the 2019 and 2015 announcement months, respectively. For expositional clarity, we present here the results of a concise difference-in-difference (DiD) estimate and relegate the event study coefficients to [Internet Appendix IX](#). Both approaches yield identical results. The DiD estimate uses an indicator variable  $post$  that equals one for the observations in the 6 months after each announcement, and another indicator variable  $I_{2019}$  that equals one for the observations in the 12 months around the 2019 announcement. The estimating equation is:

$$\begin{aligned} \Delta \log L_{b,f,t} = & \theta [post] \times QE_b^{AFS} + \lambda [post] \times QE_b^{AFS} \times [I_{2019}] \\ & + \beta QE_b^{TradingBook} \times [post] + \gamma Y_b + \delta Z_{b,t} + \psi_b + \psi_b \times [I_{2019}] \\ & + \psi_{f,t} + \varepsilon_{b,f,t}, \end{aligned}$$

where  $QE_b^{AFS}$  denotes bank  $b$ 's holding of available-for-sale securities,  $QE_b^{TradingBook}$  denotes bank  $b$ 's holding of trading book securities, and the other terms are the same as in Equation (1), for we include the same controls as in the baseline. Note, however, that we allow the bank fixed effects to vary between the 2015 and 2019 windows (i.e., we use bank-by-announcement fixed effects) to allow for the possibility that each bank's trend in loan growth rates around the 2019 announcement differs from the 2015 one. Under the assumption that only MMA securities affect banks' response to the PSPP, the coefficient  $\lambda$ —which captures the differential effect of the AFS exposure when AFS securities are valued at MMA—should be positive and statistically

**Table 4**  
**Combining the 2019 and 2015 samples: A formal test of the role of MMA**

|                                                                                           | (1)                | (2)               | (3)               | (4)                |
|-------------------------------------------------------------------------------------------|--------------------|-------------------|-------------------|--------------------|
| $[\text{post}] \times QE_b^{AFS}$                                                         | −0.017<br>[0.024]  | −0.002<br>[0.023] | −0.007<br>[0.023] |                    |
| $[\text{post}] \times QE_b^{AFS} \times [I_{2019}]$                                       | 0.095**<br>[0.041] | 0.081*<br>[0.043] | 0.081*<br>[0.041] |                    |
| $[\text{Announcement month}] \times QE_b^{AFS}$                                           |                    |                   |                   | 0.021<br>[0.037]   |
| $[\text{Announcement month}] \times QE_b^{AFS} \times [I_{2019}]$                         |                    |                   |                   | 0.148**<br>[0.074] |
| Bank-announcement FEs                                                                     | Yes                | Yes               | Yes               | Yes                |
| Firm-time FEs                                                                             | Yes                | Yes               | Yes               | Yes                |
| Size, ECB lending, reserves                                                               | Yes                | Yes               | Yes               | Yes                |
| Size $\times [I_{2019}]$ , ECB lending $\times [I_{2019}]$ , reserves $\times [I_{2019}]$ |                    | Yes               | Yes               | Yes                |
| $[\text{post}] \times QE_b^{TradingBook} \times [I_{2019}]$                               |                    |                   | Yes               | Yes                |
| Observations                                                                              | 14,214,233         | 14,214,233        | 14,214,233        | 14,214,233         |
| R-squared                                                                                 | .378               | .378              | .378              | .378               |

The dependent variable is the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  in month  $t$  (in percentages). The sample includes all observations in the 12 months around each of the 2015 and 2019 announcements.  $Post$  is dummy equal to one for each of the announcement months and the following months (i.e., September 2019 to February 2020, and January to June 2015), and  $Announcement\ month$  is a dummy equal to one for each of the announcement months (i.e., September 2019 and January 2015).  $QE_b^{AFS}$  denotes the holdings of eligible available-for-sale securities of bank  $b$  relative to total assets (in percentages), in the preannouncement month, and  $[I_{2019}]$  is a dummy equal to 1 for the observations in the 12 months around the 2019 announcement. Column 1 uses the baseline controls of Equation (1); columns 2–4 add the same set of controls interacted with the  $[I_{2019}]$  dummy; and columns 3 and 4 add the trading book exposure  $QE_b^{TradingBook}$  interacted with monthly dummies and  $[I_{2019}]$ . Standard errors are clustered at the bank level. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

significant. In addition, the coefficient  $\theta$ —which captures the baseline effect of the AFS exposure when AFS securities are valued at HCA—should not be significant. This test reflects the intuition that the only difference between 2015 and 2019 in the macroprudential accounting regime is the MMA treatment of AFS sovereign assets in 2019.

Table 4 presents the results of this nested test and confirms the role played by MMA in transmitting the effects of QE. From column 1, the coefficient  $\lambda$  of  $post \times QE_b^{AFS} \times I_{2019}$  is economically and statistically significant, implying that the impact of the 2019 PSPP on bank lending was larger among banks with greater AFS exposure, relative to the 2015 sample. A one-standard-deviation increase in AFS exposure in 2019 suggests a 0.9-percentage-point increase in lending in each of the post-announcement months. Also, the coefficient  $\theta$  on the base impact of AFS exposure (i.e., the coefficient of  $post \times QE_b^{AFS}$ ) is not significant, which is consistent with the fact that, in 2015, eligible AFS assets were valued at HCA, limiting the pass-through of the PSPP onto banks’ balance sheets.

A possible concern is that the changed economic conditions and other factors over the two time periods could influence the coefficient stability of the control variables, leading to inefficient inference. Column 2 of Table 4 thus reports a more general version of column 1, allowing the coefficients for



the control variables to differ between 2015 and 2019.<sup>26</sup> The DiD coefficient is slightly smaller in this more onerous specification but remains significant. In column 3, we build on column 2 and allow the coefficient for trading book exposure  $QE_b^{TradingBook}$  to differ between 2019 and 2015 as well by adding the additional interaction term  $[post] \times QE_b^{TradingBook} \times [I_{2019}]$ . The results continue to suggest that AFS exposure played a significant role in the transmission of QE only in 2019.

Finally, because the previous event study results in Table 3 showed that the PSPP's effect on lending is concentrated in the announcement months, column 4 replicates column 3 using a narrower DiD design that replaces the *post* indicator with a dummy equal to 1 only in the announcement months (i.e., January 2015 and September 2019). We find again no effect through AFS eligible securities when such assets are valued at HCA (i.e., in 2015), and a statistically significant effect when they are valued at MMA (i.e., in 2019). In addition, the point estimate in column 4 is much bigger than in column 3, consistent with the immediate announcement effects from Table 3. Overall, these tests confirm the mediating role of macroprudential accounting regulation in the pass-through of quantitative easing onto bank lending.

### 4.3 Robustness analyses

We conduct a long list of robustness checks for both 2019 and 2015. Below is a summary of these tests, with the details in the [Internet Appendix](#).

**4.3.1 Effects on interest rates** If the PSPP caused an expansion in credit supply, the price of credit (i.e., the interest rate on loans) should decline. In contrast, if our results reflect a coincidental increase in credit demand, interest rates should be nondecreasing in exposure to the PSPP. This section studies the impact of the PSPP on the interest rates of newly originated term loans with maturity greater than 1 year. These data are available only for a subset of banks—our final sample includes 43 banks in 2019 and 37 banks in 2015—and for firms with an overall outstanding loan balance at any given bank of at least €25,000 in 2019 and €75,000 in 2015.<sup>27</sup> The 2019 data are available at a monthly frequency, but the 2015 data are available only at a quarterly frequency.<sup>28</sup> [Internet Appendix IV](#) shows that the main results

<sup>26</sup> We include the controls for banks' size and holdings of central bank reserves (interacted with the post indicator) as well as the logarithmic change in the euro value that bank *b* borrows from the ECB at time *t*, as in our previous regressions, but we also include such regressors interacted with the dummy  $[I_{2019}]$ , which takes a value of one for observations in the 12 months around the September 2019 announcement.

<sup>27</sup> The large majority of the new term loans in our estimation sample are on the intensive margin, that is, are extended by banks to firms that were preexisting customers. The sample includes a small number of observations that refer to new lending relationships, but they represent only 0.21% of the sample in 2019 and 0.17% in 2015.

<sup>28</sup> The Bank of Italy changed the interest rate data collection process between 2015 and 2019. To make comparisons between the data collected after the changes with those collected earlier, the Bank of Italy has developed a series

**Table 5**  
**The impact of the PSPP on interest rates, 2019 and 2015**

|                                 | 2019 announcement<br>(1) |                        | 2015 announcement<br>(2) |
|---------------------------------|--------------------------|------------------------|--------------------------|
| $[2019m9] \times QE_b$          | -0.598**<br>[0.238]      | $[2015q1] \times QE_b$ | -0.971***<br>[0.351]     |
| $QE_b$ measure                  | Dummy                    |                        | Dummy                    |
| Bank FEs                        | Yes                      |                        | Yes                      |
| Firm-time FEs                   | Yes                      |                        | Yes                      |
| Size, ECB lending, and reserves | Yes                      |                        | Yes                      |
| Observations                    | 14,574                   |                        | 9,451                    |
| R-squared                       | .807                     |                        | .791                     |

The dependent variable is the interest rate on new term loans by bank  $b$  to firm  $f$  in month  $t$  (column 1) or quarter  $t$  (column 2; both in percentages). In column 1, the variable  $QE_b$  is a dummy equal to 1 if a bank is in the top tercile of the distribution of mark-to-market eligible securities over assets as of August 2019, and the results are reported relative to banks in the bottom tercile. In column 2,  $QE_b$  is a dummy equal to 1 if a bank is in the top 15% of the distribution of mark-to-market eligible securities over assets as of December 2014. Standard errors are clustered at the bank level. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

of Table 3 hold for the subsample of banks for which we have interest rate data.

We use a modified version of the baseline specification described in Equation (1) in which we continue to use firm-by-time fixed effects. The new dependent variable in Table 5 is the interest rate  $i_{b,f,t}$  on a term loan originated at time  $t$  by bank  $b$  and extended to firm  $f$ .<sup>29</sup> Because of the limited sample, we focus on the dummy measures of exposure to the PSPP to limit the effects of noise and outliers, as defined in Section 4.1. And for expositional clarity, we interact the exposure measure only with a dummy equal to one in the announcement period; Internet Appendix IV shows that the results are unchanged when we use the full set of event-study coefficients. Table 5 shows that more exposed banks reduced the interest rate in comparison to less exposed ones after the announcement. The result is statistically significant, and the magnitude is economically important: more exposed banks reduce interest rates by about 60 bps in the 2019 announcement month (column 1 of Table 5) and by 97 bps in the 2015 announcement quarter (column 2 of Table 5). This suggests that both the 2019 and 2015 PSPP announcements generated a shift in the loan supply curve, consistent with an increase in quantities and a reduction in prices.

**4.3.2 Other robustness checks.** We run several robustness checks that support our main result. This section briefly summarizes the tests that we run, and Internet Appendix II provides additional details.

of filters that can be applied to the more recent data. The filters we apply require the following to be excluded from the 2019 data: (a) loans classified as overdrafts, credit card debt, and other revolving credit; (b) loans related to trade receivables; and (c) loans awarded for the specific purpose of financing import and export activities.

<sup>29</sup> The interest rate for newly originated term loans that we use is an APR that accounts for origination fees, which is referred to as *Tasso Annuo Effettivo Globale* (TAEG) as defined by EU Council Directive 87/102/EEC.

First, we show that MMA exposure is unrelated to several key banks' characteristics. Although few bank characteristics are correlated with the continuous or dummy MMA exposure measure, the results are unchanged when controlling for such characteristics. In addition, we show that the results are unchanged when controlling for (a) holdings of noneligible securities, to rule out concerns related to possible general equilibrium effects that could affect banks through their holdings of such securities; (b) banks' exposure to other ECB policies; and (c) the comprehensive assessments conducted in 2013–2014 (i.e., stress tests and asset quality reviews).

To deal with the concern of anticipatory bias, we first note that the MMA PSPP exposure is highly persistent: the autocorrelation coefficient is 0.67 and 0.92 in the 6 months before the 2019 and 2015 announcements, respectively. We then redefine a bank as exposed if it held a large amount of MMA sovereign securities in both the preannouncement month and 6 months before, and we rerun our main regression. The results are unchanged.

To exclude the possibility that our results mechanically reflect generic differences in banks' business models that correlate with their AFS portfolios, we consider a falsification test. We measure a bank's 2015 exposure based on eligible securities classified as AFS and in the trading book—the criteria used to compute the MMA exposure in 2019. The results are insignificant.

Finally, we perform two additional tests. To deal with seasonality concerns, we expand our sample from 1 to 4 years for each announcement—with 33 million observations for the 2019 tests and 23 million for the 2015 ones—and we control for seasonal effects by including monthly dummies interacted with the PSPP exposure. The main results are again unchanged. We then perform a placebo test, repeating our main analysis 1 year before the first PSPP announcement and showing that holdings of MMA PSPP eligible securities at that time do not affect lending.

## 5. Mechanism

We have seen robust evidence that the macroprudential accounting regime mediates the impact of QE onto bank lending. This section now examines the channels of causation, focusing on the recapitalization channel, the liquidity channel, and the portfolio rebalancing channel. In a final suite of tests, we use both firm- and bank-level heterogeneity to better understand the loan supply response and the allocation of credit. Our findings show that capital gains on MMA eligible securities are the key drivers of the increase in lending, supporting the *regulatory capital* recapitalization channel.

### 5.1 Liquidity, portfolio rebalancing, and recapitalization

The first set of tests uses a triple interaction approach to examine whether banks' ex ante heterogeneity in regulatory capital or liquidity shapes their lending responses to the PSPP. These tests are related to the recapitalization

**Table 6**  
**Mechanism: Capital and liquidity**

|                                                                                               | 2019<br>announcement<br>(1) |                                                                                               | 2015<br>announcement<br>(2) |
|-----------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------|-----------------------------|
| $[2019m9] \times QE_b$                                                                        | 0.204**<br>[0.101]          | $[2015m1] \times QE_b$                                                                        | 0.692**<br>[0.286]          |
| $[2019m9] \times QE_b \times [\text{Tier 1 ratio}]_b$                                         | -0.010**<br>[0.005]         | $[2015m1] \times QE_b \times [\text{Tier 1 ratio}]_b$                                         | -0.162**<br>[0.069]         |
| $[2019m9] \times QE_b \times [\text{liquidity/assets}]_b$                                     | -0.009<br>[0.021]           | $[2015m1] \times QE_b \times [\text{liquidity/assets}]_b$                                     | 0.049<br>[0.434]            |
| $[2019m9] \times QE_b \times$<br>$[\text{Tier 1 ratio}]_b \times [\text{liquidity/assets}]_b$ | -0.003<br>[0.004]           | $[2015m1] \times QE_b \times$<br>$[\text{Tier 1 ratio}]_b \times [\text{liquidity/assets}]_b$ | 0.085<br>[0.088]            |
| Bank FEs                                                                                      | Yes                         |                                                                                               | Yes                         |
| Firm-time FEs                                                                                 | Yes                         |                                                                                               | Yes                         |
| Size, ECB lending, and reserves                                                               | Yes                         |                                                                                               | Yes                         |
| Observations                                                                                  | 8,295,389                   |                                                                                               | 5,623,310                   |
| R-squared                                                                                     | .370                        |                                                                                               | .396                        |

The dependent variable is the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  in month  $t$  (in percentages). The variable  $QE_b$  is the ratio of mark-to-market eligible securities to total assets as of August 2019 (column 1) or as of December 2014 (column 2, both in percentages). The Tier 1 ratio is the ratio of Tier 1 capital to risk-weighted assets (in percentages), and liquidity denotes central bank deposits and cash over total assets (in percentages). Tier 1 ratio and liquidity/assets are centered on their respective first quartile levels (i.e., for capital, we subtract the first quartile level of the Tier 1 capital ratio from each bank's Tier 1 ratio, and similar for liquidity). All regressions include the lower-order terms unless they are absorbed by the fixed effects. Standard errors are clustered at the bank level. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

channel and the liquidity channel, respectively, as discussed in Section 2. We use Tier 1 capital as a share of risk-weighted assets to measure banks' heterogeneity in capitalization, and central bank reserves as a fraction of total assets to measure bank-level heterogeneity in liquidity. These two variables are centered at the 25th percentile of their respective distributions, so that the results can be easily interpreted as the marginal effect relative to banks in the bottom quartile of the capital and liquidity distributions.<sup>30</sup> We then interact banks' ex ante capital with their own MMA exposure (i.e., the holdings of MMA PSPP eligible securities relative to total assets) and an indicator variable for the month of the PSPP announcement, and we do the same for liquidity. We focus on the announcement month, as the bulk of the supply response occurs at that time (see Section 4). We also consider the interaction between capital and liquidity, as both theoretical and empirical considerations suggest that banks' lending decisions might depend not just on capitalization and liquidity positions alone but also on the interplay between them.

The results, shown in Table 6, are consistent with the recapitalization channel. The coefficient for the triple interaction between a bank's Tier 1 capital ratio, MMA exposure, and announcement month dummy is negative and significant in both 2019 (column 1) and 2015 (column 2), suggesting that

<sup>30</sup> For capital, we construct the variable  $[\text{Tier 1 ratio}]_b - [\text{Tier 1 ratio}]_{25}$ , where  $[\text{Tier 1 ratio}]_{25}$  is the 25<sup>th</sup> percentile of the Tier 1 ratio distribution, and we interact this variable with the MMA exposure. We follow a similar approach for liquidity. Results are unchanged if we center capital on the regulatory 6% threshold or if we do not center capital and liquidity at all.

the impact of the PSPP on lending among MMA-exposed banks is significantly lower for more capitalized banks. Consider a bank at the median level of MMA exposure, conditional on strictly positive values. The 2019 coefficients suggest that lending increased after the announcement by about 0.83 percentage points if the bank is at the 25th percentile of the Tier 1 ratio distribution, but only 0.57 percentage points if the bank is at the 75th percentile (i.e., about 30% less). In 2015, the effects related to banks' capitalization are larger. For a bank at the 25th percentile, the impulse is 0.6 percentage points and essentially zero at the 75th percentile. That the recapitalization channel is stronger in 2015 is not surprising, given that Italian banks were less capitalized in 2015 relative to 2019 (see, for instance, the IMF Country Report No. 20/81). Of note is that we detect no effects related to banks' holdings of liquidity—neither directly nor in relation to their capital position.

While the results of Table 6 are consistent with a recapitalization channel, they could also be driven by a portfolio rebalancing channel: less capitalized banks could sell more PSPP-eligible securities to invest in loans, which have a higher expected return. We thus assess the portfolio rebalancing channel in two steps: first, we examine the impact of the PSPP on trading activity, and second, we gauge the importance of this trading activity relative to the recapitalization channel in shaping the post-announcement loan supply among more MMA-exposed banks. Note, however, that only half of this channel might operate. That is, exposed banks could sell eligible appreciated securities to hoard liquidity, reduce their borrowing from the interbank market, or reduce noncore liabilities, leaving the loan supply unchanged.

In the first step, we use the granularity of the supervisory data to measure banks' trading activity. The unit of observation is the trading of security  $s$  by bank  $b$  at time  $t$ , and a security is defined at the ISIN level—the most disaggregated.<sup>31</sup> For each bank  $b$ , we define the trading of security  $s$  in month  $t$  as the change in the quantity of security  $s$  over month  $t$  times the average market price in month  $t$ , scaled by the bank's assets in the previous month:

$$Trading_{s,b,t} = \frac{(Quantity_{s,b,t} - Quantity_{s,b,t-1}) \times Price_{s,t}^{average}}{Assets_{b,t-1}}.$$

This variable is positive if the bank is a net buyer of security  $s$  and negative if the bank is a net seller. The estimating regression uses  $Trading_{s,b,t}$  as the dependent variable and is otherwise similar to our baseline specification, but we follow Peydró, Polo, and Sette (2021) and (a) control for security-by-time fixed effects to absorb the influence of time-varying security-level factors that affect all banks equally and (b) cluster standard errors at the bank and ISIN level.<sup>32</sup>

<sup>31</sup> Each bank reports the end-of-month value of each security it owns, and we compute the end-of-month quantity using the corresponding end-of-month price.

<sup>32</sup> To construct the data set, we consider securities that were eligible to be purchased when the ECB began implementing the PSPP (i.e., those included in our exposure measure introduced in Section 4). We exclude

To better identify the role of macroprudential regulations, we analyze the trading activity separately for the trading book and available-for-sale (AFS) portfolios. In 2019, trading book and AFS securities were both valued at MMA, but in 2015, only the trading book securities were valued at MMA, as AFS securities were valued at HCA. This implies that for an AFS security with unrealized losses, such losses are not included in 2015's regulatory capital, but they would be included if a bank were to sell the security. This incentive to shield capital from unrealized AFS losses would then dampen any trading response to the 2015 PSPP announcement in the AFS portfolio. In contrast, since AFS assets were already valued at MMA in 2019, there is no major regulatory friction to trading AFS assets in response to the 2019 PSPP announcement. Thus, we should expect a bigger AFS trading response in 2019 relative to 2015.

Regarding the trading book, there should be little difference in PSPP-related trading activity when comparing 2019 with 2015. Unlike AFS securities, trading book assets were valued using MMA in both 2015 and 2019. Also, any trading book response to the PSPP might be small or insignificant, as trading book assets are rarely used for asset liability management; these assets are ordinarily used to generate trading income. Thus, any rebalancing in response to the PSPP is unlikely to be concentrated in the trading book.

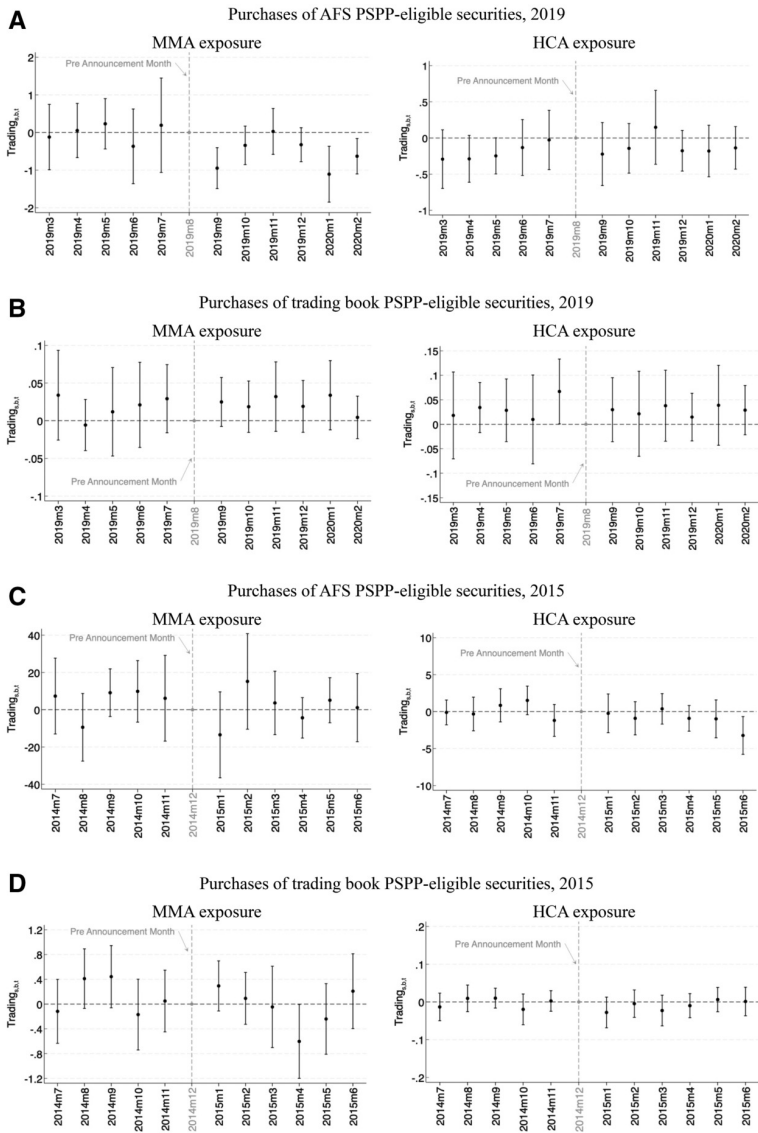
Consistent with the first step in the rebalancing channel, Figure 5 suggests that MMA exposure to the PSPP elicited a larger AFS trading response in 2019. In panel A, the dependent variable is the monthly trading of AFS PSPP-eligible securities between March 2019 and February 2020, and we regress it on both the MMA exposure interacted with monthly dummies (coefficients in the top left panel) and the HCA exposure interacted with monthly dummies (coefficients in the top right panel); the results are also reported in Table III.I, in Internet Appendix III. The effects of the HCA exposure are economically and statistically insignificant, but there is a significant impulse in trading activity after the PSPP announcement among banks with larger MMA exposures.

For September 2019, the point estimate suggests that a bank at the 75th percentile of the exposure distribution sold an additional 7.8% of its holdings of MMA eligible securities, relative to a bank at the 25th percentile. But as shown in panel C, there is no evidence that more MMA-exposed banks sold AFS securities in 2015 in line with the hypothesis discussed before.<sup>33</sup> Similarly, panels B and D show no significant trading response to the PSPP for assets held in the trading book in either 2019 or 2015. Internet Appendix III provides some

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security-time observations for which price data are not available and construct a balanced panel by assigning a zero whenever a bank does not hold a given ISIN in a given month. We also follow Peydró, Polo, and Sette (2021) and reduce the influence of securities of small value by excluding, for each bank, securities for which the average notional amount across all periods is below €10,000.

<sup>33</sup> Kojien et al. (2021) study the portfolio rebalancing triggered by the PSPP between 2015 and 2017, and they find relatively small sales by banks, despite their large holdings. Our portfolio rebalancing results can explain the limited sales by banks, as HCA was prevalent during the period they analyze.



**Figure 5**  
**The impact of the PSPP on bank trading activity**

The figure presents the result of a series of regressions in which the dependent variable is the purchase in EUR (or sale if negative) of security  $s$  of bank  $b$  in month  $t$  per EUR millions of total assets in  $t - 1$ , where a security is defined at the ISIN level. Panels A and B use a 12-month window around the 2019 announcement. Panel A refers to one single specification in which  $s$  is a PSPP-eligible security in the AFS portfolio and the specification includes two measures of exposure: one based on MMA PSPP-eligible securities and the other based on HCA PSPP-eligible securities. Panel B refers to one single specification in which  $s$  is a PSPP-eligible security in the trading book, and the specification includes two measures of exposure: one based on MMA PSPP-eligible securities and the other based on HCA PSPP-eligible securities. Panels C and D follow an identical pattern for the 2015 announcement. All regressions include bank fixed effects, security-time fixed effects, and the baseline controls (i.e., size, ECB lending, and reserves). Standard errors are clustered at the bank and security level. The bars represent the 90% confidence interval around each point estimate.

robustness checks for these results—using the MMA exposure dummy and alternative trading measures—and some additional analysis of banks' trading activity.<sup>34</sup>

We have seen evidence that banks more exposed to the 2019 PSPP announcement increased their trading of eligible AFS securities. While this first step is consistent with the rebalancing channel, banks might not have used the proceeds of the sales to increase lending. We now evaluate the relative importance of any PSPP-related rebalancing in determining the lending response of banks. To compare the portfolio rebalancing and recapitalization channels, we construct bank-level measures of capital gains on MMA eligible securities as well as bank-level measures of trading revenues from eligible securities in a narrow 1-month window around each PSPP announcement. These two measures capture the *regulatory capital* recapitalization channel (i.e., the recapitalization channel that affects regulatory capital) and the portfolio rebalancing channel, respectively.

We measure banks' capital gains on MMA eligible securities as the product of the lagged quantity and the change in price, relative to total assets:

$$CapitalGains_{b,t} = \frac{\sum_s (Quantity_{s,b,t-1} \times (Price_{s,t} - Price_{s,t-1}))}{Assets_{b,t-1}}.$$

Likewise, the bank-level measure of trading revenues is computed by adding up the ISIN-level measure  $Trading_{s,b,t}$  introduced before:<sup>35</sup>

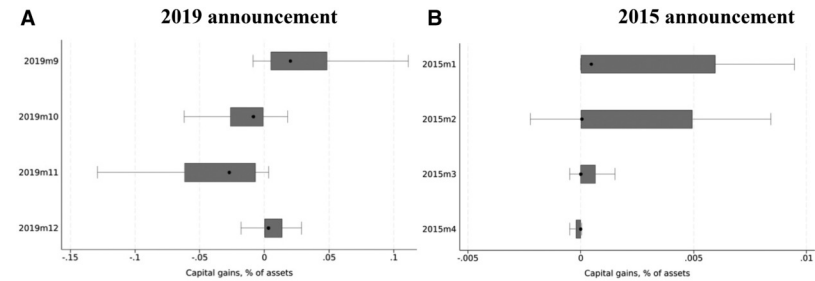
$$Trading_{b,t} = \frac{\sum_s (Quantity_{s,b,t} - Quantity_{s,b,t-1}) \times Price_{s,t}^{average}}{Assets_{b,t-1}}.$$

The variable  $Trading_{b,t}$  is constructed using eligible securities in the AFS and trading book portfolio, but [Internet Appendix III.4](#) shows that the results are unchanged if we restrict attention to only one portfolio or the other, or if we consider all securities, including those that are not eligible to be purchased under the PSPP. Both  $CapitalGains_{b,t}$  and  $Trading_{b,t}$  are computed in the month immediately upon the PSPP's announcements, and because of the way they are defined, they measure capital gains and trading activity relative to the month just before. These two measures thus capture the cross-sectional variation in banks' capital gains and revenues from trading activity just around the two PSPP announcements, respectively. Figures 6 and 7 plot the distribution

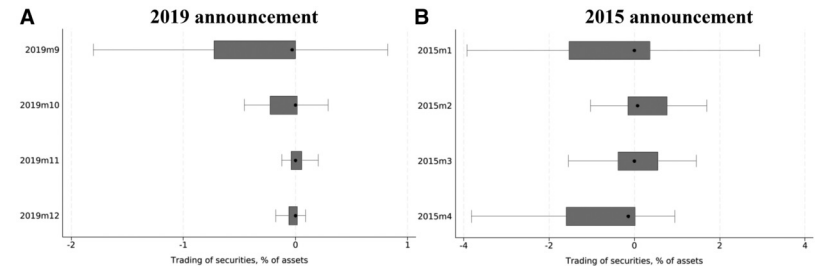
<sup>34</sup> In the additional analysis conducted in [Internet Appendix III.2](#) and [III.3](#), we link the trading activity to banks' capital and liquidity position, and we show that the PSPP announcement essentially has no effect on the trading of securities that are not eligible to be purchased by the ECB under the PSPP.

<sup>35</sup> Because we are constructing bank-level measures of trading activity (as opposed to security-level measures, as we did before), we now consider all eligible securities, including those with a notional amount below €10,000, which we omitted from the security-level analysis. Also note that because price data are not available for some security-time observations, we remove such observations from the computation of both capital gains and trading, to make sure that the lack of price data does not affect the horse race between capital gains and trading.





**Figure 6**  
**Evolution of capital gains after the PSPP announcement**  
The figure presents box plots of the evolution of capital gains in the month of the PSPP announcement and in the following 3 months, for banks with strictly positive holdings of MMA-eligible securities in the preannouncement month. We plot the median (black dot), the 25th/75th percentiles (lower/upper ends of box), and the lower/upper adjacent values (lower/upper limits on whiskers). The lower (upper) adjacent value is the furthest observation within one and a half interquartile ranges of the lower (upper) end of the box.



**Figure 7**  
**Evolution of trading after the PSPP announcement**  
The figure presents box plots of the evolution of the variable  $Trading_{b,t}$  in the month of the PSPP announcement and in the following 3 months, for banks with strictly positive holdings of MMA-eligible securities in the preannouncement month. We plot the median (black dot), the 25th/75th percentiles (lower/upper ends of box), and the lower/upper adjacent values (lower/upper limits on whiskers). The lower (upper) adjacent value is the furthest observation within one and a half interquartile ranges of the lower (upper) end of the box.

of capital gains and revenues from trading activity in the announcement month and the following few months, respectively, confirming that most of the effect was immediate.

The logic of the reduced-form tests is that if the portfolio rebalancing channel operates, lending should increase more among those banks with the most trading activity after the PSPP. And if the recapitalization channel operates, lending should increase the most among banks that experience the largest capital gains in their MMA securities, even after conditioning on the trading response. To implement this test, as before, we use the credit register data to construct bank-firm lending growth as the dependent variable, similar to Equation (1). In the case of the 2019 PSPP announcement, the specification then interacts both  $CapitalGains_{b,t}$  and  $Trading_{b,t}$  with the September 2019

Table 7  
Capital gains versus trading

|                                                                    | 2019 announcement   |                      |                                                                    | 2015 announcement    |                      |
|--------------------------------------------------------------------|---------------------|----------------------|--------------------------------------------------------------------|----------------------|----------------------|
|                                                                    | (1)                 | (2)                  |                                                                    | (3)                  | (4)                  |
| [2019m9] × [(Trading 2019m9)/Assets] <sub>b</sub>                  | −0.135<br>[0.184]   | −0.267<br>[0.188]    | [2015m1] × [(Trading 2015m1)/Assets] <sub>b</sub>                  | 0.067<br>[0.087]     | 0.069<br>[0.088]     |
| [2019m9] × [(Capital gains 2019m9)/Assets] <sub>b</sub>            | 19.667**<br>[9.181] |                      | [2015m1] × [(Capital gains 2015m1)/Assets] <sub>b</sub>            | 44.500***<br>[7.823] |                      |
| [2019m9] × [(Unrealized capital gains 2019m9)/Assets] <sub>b</sub> |                     | 22.031**<br>[10.954] | [2015m1] × [(Unrealized capital gains 2015m1)/Assets] <sub>b</sub> |                      | 51.837***<br>[8.042] |
| Bank FEs                                                           | Yes                 | Yes                  |                                                                    | Yes                  | Yes                  |
| Firm-time FEs                                                      | Yes                 | Yes                  |                                                                    | Yes                  | Yes                  |
| Size, ECB lending, and reserves                                    | Yes                 | Yes                  |                                                                    | Yes                  | Yes                  |
| Observations                                                       | 8,346,925           | 8,346,925            |                                                                    | 5,867,308            | 5,867,308            |
| R-squared                                                          | .369                | .369                 |                                                                    | .394                 | .394                 |

The dependent variable is the change in the logarithm of disbursed loans from bank *b* to firm *f* in month *t* (in percentages). *Capital gains* are measured as the product of lagged quantities of marked-to-market PSPP-eligible securities and the change in price, and they are scaled by assets in the preannouncement month (in percentages). *Trading* is measured as the net purchases of PSPP-eligible securities in the AFS and trading book portfolios, scaled by assets in the preannouncement month (in percentages). *Unrealized capital gains* are the *capital gains* for the securities that are not sold; that is, for each security *s*, if the bank had a long position, we include the capital gains on the amount that is not sold, and if the bank had a short position, we include the capital gains on the part of the position that remains open; for positions that switch from long to short or vice versa, we assign zero capital gains. Standard errors are clustered at the bank level. \**p* < .1; \*\**p* < .05; \*\*\**p* < .01.

indicator variable, using the same controls and fixed effects as in our baseline specification. The 2015 specification is analogous.

Column 1 of Table 7 reports the results of this test for the 2019 PSPP announcement. The evidence suggests that the recapitalization of the banking system remains the key channel through which the PSPP increased lending. A one-standard-deviation increase in capital gains in a bank’s MMA portfolio between August and September 2019 is associated with an increase in lending of 2.36 percentage points. We find no evidence that trading activity over the same narrow window is associated with any increase in lending.

A possible concern is that capital gains might affect banks’ behavior differently depending on whether the underlying securities are kept or sold (i.e., whether the capital gains are realized or unrealized). That is, trading might still have an indirect effect by creating a differential response between realized and unrealized capital gains. To address this issue, column 2 of Table 7 repeats the analysis of column 1 by calculating only the capital gains on eligible securities that are not sold.<sup>36</sup> The results, however, are essentially the same as in column 1.

<sup>36</sup> More precisely, for a long position, we include only capital gains on the securities that are not sold, and for short positions that are partially closed, we include only the capital gains on the part of the position that remains open. For positions that switch from long to short or vice versa, we assign zero capital gains.

Columns 3 and 4 repeat the exercises for the 2015 PSPP. The results are similar, and again we observe no difference regardless of whether we use all capital gains or only unrealized capital gains. Lending increased the most among those banks with the largest capital gains in the 1-month window around the 2015 PSPP announcement. In this case, a one-standard-deviation increase in capital gains suggests a 0.45-percentage-point increase in lending. And as in 2019, the coefficient for trading activity over the 2015 window is insignificant.

Finally, to formalize the result that the effects of the PSPP through capital gains were mostly immediate upon the announcement, we build on the approach used in Table 7 and interact monthly dummies with the respective measure of capital gain (as a fraction of assets), for each post-announcement month. That is, for the 2019 event, we use the September indicator interacted with the September capital gains, the October indicator interacted with the October capital gains, and so on, and similarly for 2015. The results are represented in Figure 8.<sup>37</sup> For each month, the coefficient represents the effect on lending (in percentage points) of increasing capital gains by one standard deviation. In both 2019 and 2015, capital gains affect lending only in the announcement month and do not have an impact in the following months, a finding that is consistent with the observation in Figure 6 that capital gains are concentrated in the announcement month. Internet Appendix III.6 repeats the analysis by adding the bank-level measure of trading interacted with monthly dummies, showing that trading has no impact on lending in the announcement month (as shown in Table 7) or in all other post-announcement months.

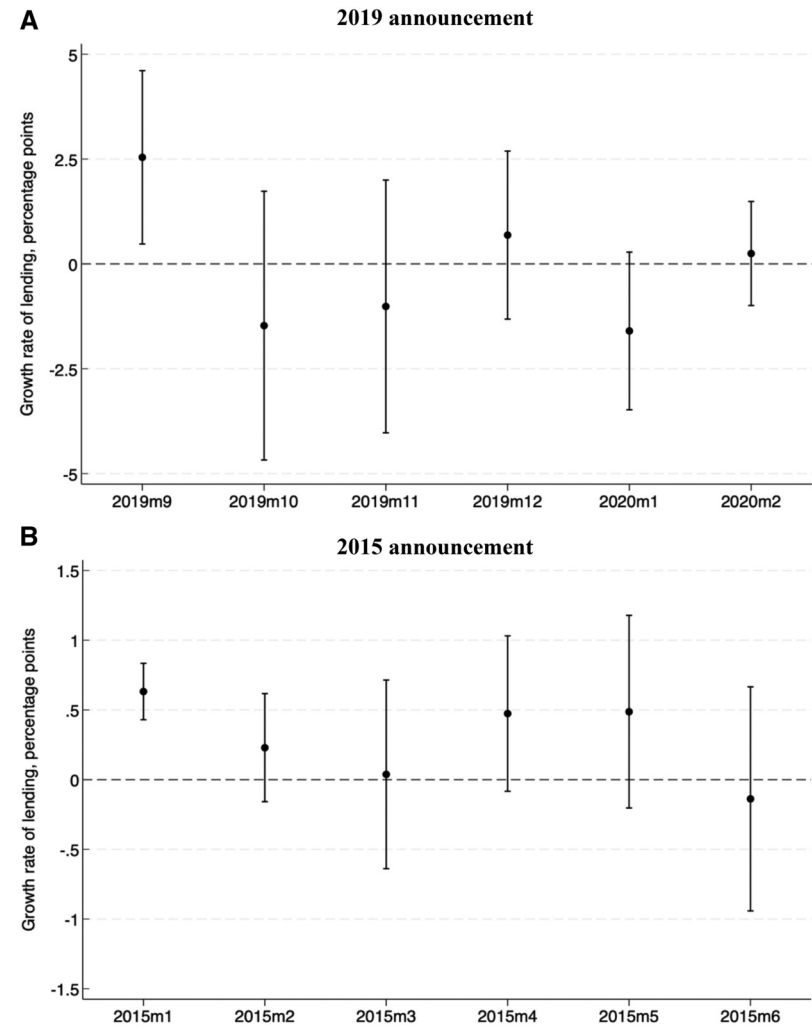
We provide a series of robustness checks and additional tests in Internet Appendix III.4 and III.5. In particular, Internet Appendix III.5 argues that the recapitalization induced by the capital gains is large enough to justify the increase in lending volume immediately after the announcement, and that the effects of the PSPP are essentially transmitted only through capital gains. That is, after accounting for capital gains, the cross-sectional variation in banks' holdings of PSPP-eligible securities valued at MMA has no explanatory power on bank lending.

## 5.2 The allocation of credit

The evidence based on bank-level heterogeneity suggests that the PSPP increased lending mainly via the recapitalization channel and, in particular, by increasing banks' regulatory capital. This section combines bank- and firm-level heterogeneity to study the allocation of credit in the wake of the two PSPP announcements.<sup>38</sup> A key objective is to assess what role, if any, the macroprudential accounting regime played in shaping banks' risk-taking behavior in the corporate loan book.

<sup>37</sup> Table III.H in Internet Appendix III lists the estimated coefficients.

<sup>38</sup> Internet Appendix VII provides another analysis related to the allocation of credit by studying whether more exposed banks respond to the PSPP by increasing lending not only to preexisting customers but also to new customers.



**Figure 8**  
**The effects of capital gains on lending**

The figure presents the results of a regression for each announcement, in which the dependent variable is the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  in month  $t$  (in percentages) in the 12-month window around the announcement. Each panel plots the coefficients of  $[capital\ gains]_{b,t}/[assets]_{b,t-1}$  interacted with an indicator equal to one in month  $t$ , for each post-announcement month. For each month  $t$ , the coefficient represents the effect on lending (in percentage points) of increasing capital gains in month  $t$  by one standard deviation. The regressions include bank fixed effects, firm-time fixed effects, and the baseline controls (i.e., size, ECB lending, and reserves). Standard errors are clustered at the bank level. The bars represent the 90% confidence interval around each point estimate.

The “risk-taking channel” predicts heterogeneity in the types of firms that obtain credit after expansionary policies like the PSPP (Bauer, Bernanke, and Milstein 2023; Kashyap and Stein 2023). This channel notes that expansionary policies can increase the risk appetite of financial intermediaries, increasing loan supply to riskier firms. But the risk-taking channel also notes that bank-level heterogeneity can shape the allocation of credit across firm types. For example, monetary stimulus can help undercapitalized banks derisk, so that these banks might lend less to riskier firms after the PSPP (Pegoraro 2024; Peydró, Polo, and Sette 2021). Regulatory considerations might also affect the risk-taking response of banks, as riskier loans might require banks to lock in more capital. Thus, less capitalized banks might prefer to lend to less risky firms to take more advantage of the ability to expand lending. Bank-level heterogeneity can also shape the allocation of credit across firm types.

Building on these arguments, Table 8 examines the allocation of credit after the PSPP announcements. We modify the baseline specification of Equation (1) by interacting measures of firms’ type with a bank’s MMA exposure around the PSPP’s announcement. Because this approach results in a series of triple interaction terms and their lower order counterparts, which greatly multiply the number of parameters to be estimated, we focus again on the credit supply response in the month of the announcement. That is, we interact the firm’s type with the bank’s MMA exposure along with an indicator variable for the month of the PSPP announcement. As shown in the previous sections, much of the lending response was immediate, and little information is lost with this approach.

We begin with the 2019 analysis, the results of which are reported in the top panel of Table 8. Because firm-level data are available only for a subset of the firms included in the baseline regression, column 1 reproduces the baseline result using this subsample—and the more concise specification. We again find that lending disproportionately increased in the month of the PSPP’s announcement among MMA-exposed banks even within the restricted sample.

Column 2 considers the role of firm risk in the allocation of credit, adding the interaction with an indicator variable that equals one for risky firms. Credit risk is measured on a 0–10 scale, with higher scores denoting a riskier firm, and we set our indicator variable to one for firms with a score greater than or equal to 7, and zero otherwise; about 15% of the firms in our sample are flagged as risky. The interaction term is negative and significant, implying that riskier firms received less bank credit after the PSPP announcement. A bank at the median MMA exposure increased lending by 0.71 percentage points to nonrisky firms but only by 0.12 percentage points to risky firms. Column 3 adds an interaction term with firm size, which is an indicator variable that equals 1 if the firm has revenues that exceed €50 million, and zero otherwise. The result that MMA-exposed banks lent to safer firms in the wake of the 2019 PSPP is unchanged, and column 3 also shows that such banks directed their lending more toward smaller firms.

**Table 8**  
**The allocation of credit**  
 2019 PSPP announcement

| Variables                                                              | (1)                | (2)                | (3)                 | (4)                 | (5)                | (6)                |
|------------------------------------------------------------------------|--------------------|--------------------|---------------------|---------------------|--------------------|--------------------|
| [2019m9] $\times Q E_b$                                                | 0.199**<br>[0.093] | 0.226**<br>[0.102] | 0.258**<br>[0.107]  | 0.235**<br>[0.105]  |                    |                    |
| [2019m9] $\times Q E_b \times$ [Risky firm]                            |                    | -0.189*<br>[0.100] | -0.209**<br>[0.100] | -0.204**<br>[0.096] | -0.164*<br>[0.092] | -0.153*<br>[0.089] |
| [2019m9] $\times Q E_b \times$ [Big firm]                              |                    |                    | -0.217*<br>[0.125]  |                     | -0.196<br>[0.138]  |                    |
| [2019m9] $\times Q E_b \times$ [Risky firm]<br>$\times$ [Tier 1 ratio] |                    |                    |                     | 0.014**<br>[0.006]  |                    | 0.009<br>[0.006]   |
| Observations                                                           | 4,911,812          | 4,911,812          | 4,911,812           | 4,911,812           | 4,911,812          | 4,911,812          |
| R-squared                                                              | .348               | .348               | .348                | .348                | .351               | .351               |
| Bank FEs                                                               | Yes                | Yes                | Yes                 | Yes                 |                    |                    |
| Bank-time FEs                                                          |                    |                    |                     |                     | Yes                | Yes                |
| Firm-time FEs                                                          | Yes                | Yes                | Yes                 | Yes                 | Yes                | Yes                |
| Size, ECB lending, reserves                                            | Yes                | Yes                | Yes                 | Yes                 | Yes                | Yes                |

2015 PSPP announcement

| Variables                                                              | (1)                 | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|------------------------------------------------------------------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| [2015m1] $\times Q E_b$                                                | 0.523***<br>[0.108] | 0.733***<br>[0.129]  | 0.771***<br>[0.109]  | 1.236**<br>[0.493]   |                      |                      |
| [2015m1] $\times Q E_b \times$ [Risky firm]                            |                     | -0.597***<br>[0.191] | -0.624***<br>[0.172] | -1.635***<br>[0.496] | -0.630***<br>[0.164] | -1.636***<br>[0.521] |
| [2015m1] $\times Q E_b \times$ [Big firm]                              |                     |                      | -0.489<br>[0.772]    |                      | -0.401<br>[0.819]    |                      |
| [2015m1] $\times Q E_b \times$ [Risky firm]<br>$\times$ [Tier 1 ratio] |                     |                      |                      | 0.214**<br>[0.091]   |                      | 0.215**<br>[0.098]   |
| Observations                                                           | 2,855,099           | 2,855,099            | 2,855,099            | 2,855,099            | 2,855,099            | 2,855,099            |
| R-squared                                                              | .378                | .378                 | .378                 | .378                 | .380                 | .380                 |
| Bank FEs                                                               | Yes                 | Yes                  | Yes                  | Yes                  |                      |                      |
| Bank-time FEs                                                          |                     |                      |                      |                      | Yes                  | Yes                  |
| Firm-time FEs                                                          | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Size, ECB lending, reserves                                            | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |

The dependent variable is the change in the logarithm of disbursed loans from bank  $b$  to firm  $f$  in month  $t$  (in percentages).  $Q E_b$  denotes the holdings of PSPP-eligible securities valued at MMA, relative to total assets (in percentages). *Risky firm* is a dummy equal to 1 for firms with high credit risk, and *Big firm* is a dummy equal to 1 for firms with revenues exceeding €50 million; both firm-level variables are as of the end of the previous calendar year. The *Tier 1 ratio* is the ratio of Tier 1 capital to risk-weighted assets (in percentages), centered on the first quartile value of its distribution. All regressions include lower-order interaction terms. Standard errors are clustered at the bank level. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

We have seen evidence that the PSPP increased lending via the recapitalization channel, and column 4 of Table 8 combines bank- and firm-level heterogeneity to examine whether banks' ex ante regulatory capital also shaped risk-taking incentives. To this end, we use a quadruple interaction term. We interact a bank's MMA exposure with the PSPP announcement month dummy, the indicator variable that equals 1 if a firm is high risk, and the bank's Tier 1 capital ratio (i.e., the ratio of Tier 1 capital to risk-weighted assets). We include the lower-order interactions as well. The quadruple term is positive and significant, suggesting that the most undercapitalized MMA-exposed banks

lent relatively less to riskier firms after the 2019 PSPP announcement—riskier banks derisked the most.

Consider a bank that is at the median MMA exposure and at the 75th percentile of the Tier 1 ratio distribution. For such a bank, the loan supply to nonrisky firms grew 0.37-percentage-points more than the loan supply to risky firms. For an otherwise identical but less capitalized bank—at the 25th percentile of the Tier 1 ratio distribution—the gap is wider at 0.64 percentage points. The result about the higher derisking by less capitalized banks might appear at odd with some of the literature that analyses banks' response to other unconventional ECB interventions. For instance, Acharya et al. (2019) show that the Outright Monetary Transactions (OMT) program recapitalized banks, similar to our finding, but weakly capitalized banks shifted their supply toward low-quality borrower. We note, however, that between the launch of the OMT in 2012 and the PSPP announcement in 2015, many banks went through a comprehensive assessment (i.e., asset quality review and stress tests) that forced them to improve their capital position.<sup>39</sup> Thus, Italian banks in our sample had better capital positions than a few years earlier, when the OMT was launched.

The remaining columns of Table 8 exploit the firm-level heterogeneity to use bank-by-time fixed effects as an additional robustness check.<sup>40</sup> The evidence continues to suggest that banks more exposed to the PSPP lent less to riskier firms after the PSPP (column 5), though the quadruple interaction term in column 6 is less precisely estimated.

Finally, the bottom panel of Table 8 replicates the 2019 exercise for the 2015 PSPP announcement. The results are similar to those derived for 2019: the PSPP has a weaker impact on the credit supplied to riskier firms, derisking is relatively bigger among the less capitalized banks, and the results are robust to including bank-by-time fixed effects.

## 6. Extension: Other Monetary Policy Shocks

This section examines whether our results extend to other monetary policy tools in different circumstances. A possible concern is that our results are based on a specific QE policy—the PSPP—which, despite being the largest ECB asset purchases program, was implemented in response to specific economic circumstances. That is, the concern is that the results may not generalize to other monetary tools, such as forward guidance, that also aim to influence

<sup>39</sup> See [www.bankingsupervision.europa.eu/banking/tasks/comprehensive\\_assessment/html/2014\\_index.en.html](http://www.bankingsupervision.europa.eu/banking/tasks/comprehensive_assessment/html/2014_index.en.html) and [https://www.bancaditalia.it/media/comunicati/documenti/2014-02/cs\\_261014\\_en.pdf?language\\_id=1](https://www.bancaditalia.it/media/comunicati/documenti/2014-02/cs_261014_en.pdf?language_id=1).

<sup>40</sup> These nonparametric controls absorb the MMA exposure interacted with the indicator of the PSPP announcement month.

the longer end of the yield curve. This section, however, shows that the pass-through of any monetary policy shock to the long-end of the yield curve—not just QE—is mediated by the macroprudential accounting regime.

To gauge the generalizability of our results, we broaden the estimation window to span the period July 2010—when the macroprudential accounting rules switch to HCA for available-for-sale (AFS) securities—through February 2020—just before the COVID-19 pandemic.<sup>41</sup> We then examine the effects of monetary policy shocks to the long end of the yield curve—so-called “*path* rate shocks”—on bank loan supply. We use shocks from the Euro Area Monetary Policy Event-Study Database (EA-MPD, Altavilla et al. 2019), which are extracted from asset prices in 10-minute windows bracketing the ECB’s regular release of monetary policy information after each of its 10 or so meetings per year. By examining the high-frequency change in bond yields around these announcements, the EA-MPD shocks can help us measure the loan supply responses that are free of anticipatory biases. The EA-MPD also reports shocks to the short end of the yield curve—*target* rate shocks—but during our sample period, the policy or target rate was at the zero lower bound, and most target rate shocks were very small (Figure 9, dotted line). In contrast, shocks to the long end of the yield curve were often sizeable, as forward guidance and other monetary policy interventions altered longer-term yields even while the policy rate was at the zero lower bound. We use shocks to 10-year yields on Italian bonds (Figure 9, solid line), as Italian bonds represent the large majority of sovereign assets held by Italian banks.

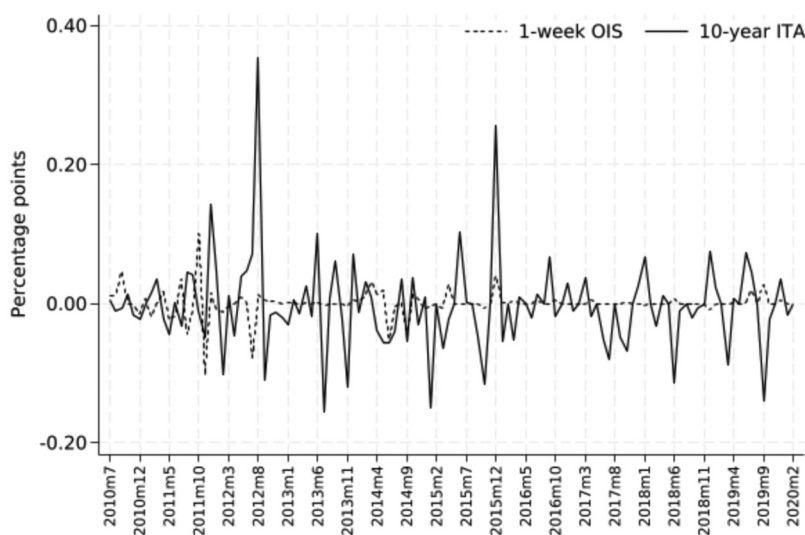
Table 9 studies the impact of path rate shocks on loan supply using the monthly credit register data. We use the specification

$$\Delta \log L_{b,f,t} = \beta IT10_t \times \left( \frac{\text{SovereignMMA5+}}{\text{Assets}} \right)_{b,t-1} + \gamma Y_{b,t-1} + \delta Z_{b,t} + \psi_b + \psi_{f,t} + \varepsilon_{b,f,t}.$$

Our dependent variable is the monthly loan growth rate at the bank-firm level, as in our baseline specification. We then use a monthly time series  $IT10_t$  of the path rate shocks—shocks to 10-year Italian bond yields in the window around the ECB’s statement and press conference—and interact this series with a bank’s 1-month-lagged sovereign MMA exposure, scaled by the bank’s assets. Note that because we consider path rate shocks on 10-year Italian bonds (i.e., on long-term yields), we measure MMA exposure as the value of sovereign bonds on a bank’s balance sheet with a residual maturity of at least 5 years

<sup>41</sup> Regarding the choice of the sample period, we note that before July 2010, the accounting rule used a mix of HCA and MMA for AFS securities, allowing only a partial pass-through of unrealized capital gains and losses to regulatory capital. In addition, we exclude the COVID-19 crisis and the subsequent years because of the various policies that have been implemented and that directly targeted bank lending. For instance, the Italian government offered a guarantee on a large fraction of long-term bank loans, and this policy is still influencing the dynamic of bank lending as of the writing of this paper.





**Figure 9**  
**Monetary policy shocks to short- and long-term rates, 2010–2020**

The figure plots the monetary policy shocks from the Euro Area Monetary Policy Event-Study Database (EA-MPD, Altavilla et al. 2019) for the period July 2010 to February 2020. These shocks are extracted from movements in market yields in a narrow window around ECB's regular release of monetary policy information after its meetings. The dotted line represents the shocks to the 1-week overnight index swap rate, and the solid line represents the shocks to 10-year Italian bonds.

(i.e., long-term sovereign bonds) that are subject to MMA. As before, we employ firm-by-month fixed effects to absorb latent demand as well as bank fixed effects, along with the other suite of controls employed in our baseline analysis.<sup>42</sup>

Consistent with the idea that a decline in long-term rates can increase MMA-based asset valuations, banks' regulatory capital, and with it, bank lending, column 1 of Table 9 shows that the interaction between the path rate shock series  $IT10_t$  and a bank's MMA exposure is negative and significant. This result suggests that a negative monetary policy shock to the longer end of the yield curve positively affects bank lending among MMA-exposed banks. In response to a 1-percentage-point negative shock, a bank in the median of the 2010–2017 sovereign MMA exposure—when MMA was restricted to the trading book—increases lending by 0.08%.<sup>43</sup> However, for a bank at the median of the 2018–2020 exposure—when MMA was extended to available-for-sale (AFS) securities—the increase in lending is 2.65%.

<sup>42</sup> To adapt the baseline specification to the study of monetary policy shocks, we control for bank size in month  $t-1$  (using log assets at  $t-1$ ) and for banks' holdings of central bank reserves in month  $t-1$ .

<sup>43</sup> More precisely, we consider the median of the sovereign MMA exposure conditional on positive MMA holdings.

**Table 9**  
**Other monetary policy shocks (July 2010 to February 2020)**

|                                                                    | (1)                     | (2)                     | (3)                     | (4)                     | (5)                     |
|--------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|                                                                    | $\Delta \log L_{b,f,t}$ | $\Delta \log L_{b,f,t}$ | $\Delta \log L_{b,f,t}$ | $\Delta \log L_{b,f,t}$ | $\Delta \log L_{b,f,t}$ |
| [IT10] <sub>t</sub>                                                | -1.170***               | -1.215***               | -0.909**                |                         | -1.296***               |
| × [Sovereign MMA (5+)/Assets] <sub>b,t-1</sub>                     | [0.339]                 | [0.358]                 | [0.395]                 |                         | [0.488]                 |
| [IT2] <sub>t</sub>                                                 |                         | 0.163                   |                         |                         |                         |
| × [Sovereign MMA (<5)/Assets] <sub>b,t-1</sub>                     |                         | [0.259]                 |                         |                         |                         |
| [IT10] <sub>t</sub>                                                |                         |                         |                         | 0.138                   |                         |
| × [Sovereign AFS (5+)/Assets] <sub>b,t-1</sub>                     |                         |                         |                         | [0.128]                 |                         |
| [IT10] <sub>t</sub> [Sovereign AFS (5+)/Assets] <sub>b,t-1</sub>   |                         |                         |                         | -0.985**                |                         |
| × [post-January 2018]                                              |                         |                         |                         | [0.487]                 |                         |
| [IT10] <sub>t</sub>                                                |                         |                         |                         | -1.346***               |                         |
| × [Sovereign trading book (5+)/Assets] <sub>b,t-1</sub>            |                         |                         |                         | [0.423]                 |                         |
| [IT10] <sub>t</sub>                                                |                         |                         |                         | 0.360                   |                         |
| × [Sovereign held to maturity (5+)/Assets] <sub>b,t-1</sub>        |                         |                         |                         | [0.290]                 |                         |
| [IT10] <sub>t</sub> × [Sovereign MMA (5+)/Assets] <sub>b,t-1</sub> |                         |                         |                         |                         | 0.032***                |
| × [Tier 1 ratio] <sub>b,year-1</sub>                               |                         |                         |                         |                         | [0.011]                 |
| [IT10] <sub>t</sub> × [Sovereign MMA (5+)/Assets] <sub>b,t-1</sub> |                         |                         |                         |                         | -0.054                  |
| × [Liquidity/Assets] <sub>b,t-1</sub>                              |                         |                         |                         |                         | [0.209]                 |
| [IT10] <sub>t</sub> × [Sovereign MMA (5+)/Assets] <sub>b,t-1</sub> |                         |                         |                         |                         | 0.001                   |
| × [Tier 1 ratio] <sub>b,year-1</sub>                               |                         |                         |                         |                         | [0.009]                 |
| × [Liquidity/Assets] <sub>b,t-1</sub>                              |                         |                         |                         |                         |                         |
| Bank FEs                                                           | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Firm-time FEs                                                      | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Size, ECB lending, and reserves                                    | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Observations                                                       | 51,461,043              | 51,461,043              | 50,575,406              | 51,461,043              | 51,461,043              |
| R-squared                                                          | .402                    | .402                    | .402                    | .402                    | .402                    |

The dependent variable is the change in the logarithm of disbursed loans from bank *b* to firm *f* in month *t* (in percentages). The sample period goes from July 2010 to February 2020 and excludes January 2018 (i.e., the month in which the macroprudential accounting rule switched to MMA for available-for-sale securities). *IT10* is the monetary policy shock for 10-year Italian bonds (in percentage points) from the EA-MPD (Altavilla et al. 2019). *Sovereign MMA (5+)* is the amount of mark-to-market sovereign securities with maturity of at least 5 years, and *Sovereign MMA (<5)* is the amount of sovereign securities with maturity of less than 5 years. *Sovereign AFS (5+)* is the amount of sovereign securities with maturity of at least 5 years in the available-for-sale portfolio; *Sovereign trading book (5+)* and *Sovereign held to maturity (5+)* are defined similarly. *Post-January 2018* is an indicator equal to 1 after January 2018, and zero otherwise. The *Tier 1 ratio* is the ratio of Tier 1 capital to risk-weighted assets at the end of the previous year (in percentages), and *Liquidity* is the sum of central bank deposits and cash at the end of the previous month. All variables that are defined as a fraction of total assets are in percentages. The Tier 1 ratio and liquidity/assets are centered on their respective first quartile levels. All regressions include the lower-order terms unless they are absorbed by the fixed effects. Standard errors are clustered at the bank level. \**p* < .1; \*\**p* < .05; \*\*\**p* < .01.

The value of shorter-duration bonds is less sensitive to interest rate movements, and shorter-term interest rates did not move much in our sample period, as noted before. In keeping with these facts, column 2 shows that holdings of sovereign bonds with a maturity less than 5 years do not trigger any sizeable response of loan supply in response to shorter-term path rate shocks—those derived from the 2-year Italian government bond—while the effect of 10-year rate shocks remains unchanged.

Because the PSPP announcements are sizeable shocks to the long end of the yield curve, column 3 drops the 2015 and 2019 PSPP announcement months from the sample period to isolate the effect of other shocks. The point estimate remains significant and suggests that shocks other than QE to the long end of the yield curve can engender a sizeable loan supply response, too.

The remaining columns build on our previous tests to deepen our understanding of the role of MMA in transmitting these more general monetary policy shocks onto bank loan supply decisions. To do this, column 4 of Table 9 creates a *post-January 2018* dummy variable that equals 1 in the months after regulators switched from HCA to MMA to value AFS sovereign assets (i.e., after January 2018), and 0 otherwise. We then interact this dummy variable with the lag ratio of AFS sovereign assets to total bank assets and the monetary policy shock series  $IT10_t$ . We also include all lower-order interaction terms. If MMA helps transmit these shocks onto bank lending, then AFS sovereign assets should affect lending decisions after a shock only when those assets are valued using MMA. That is, the coefficient for the interaction  $IT10_t \times AFS$  should be negative and significant only when interacted with the *post* dummy. Column 4 also interacts the monetary policy shock series  $IT10_t$  with bond holdings that were always valued at MMA throughout the sample period (i.e., those in the trading book) and separately with those holdings that were always valued using HCA (i.e., those held to maturity), all scaled by bank assets.

Consistent with the previous evidence based on the PSPP, the coefficient for  $IT10_t \times AFS \times post$  is negative and significant, but the coefficient for  $IT10_t \times AFS$  is not significant. The coefficient that refers to trading book assets—always valued at MMA—is also negative and significant, and the coefficient for held-to-maturity assets—always valued at HCA—is not significant. This suggests that the relative size of a bank's MMA assets, possibly through the regulatory capital recapitalization channel, is an important element of the monetary transmission mechanism.

Consistent with the salience of the regulatory capital recapitalization channel, column 5 returns to the baseline case (column 1) and shows that the impact of monetary policy shocks on lending via MMA exposure is significantly weaker for the better-capitalized banks.<sup>44</sup> For a bank at the median MMA exposure level (conditional on positive MMA holdings), a 1-percentage-point negative shock is associated with a 0.41-percentage-point increase in lending if the bank is also at the 25th percentile of Tier 1 capital to risk weighted assets, but 0.29 percentage points if the bank is at the 75th percentile (i.e., almost 30% lower). And similar to Section 5.1, we find no effects of the liquidity position of the bank.

These results (and a few robustness tests in Internet Appendix VIII) suggest that the transmission onto bank lending of other monetary policy shocks that

<sup>44</sup> For each observation in our sample, we measure capitalization using the Tier 1 ratio at the end of the previous year, centered on the first quartile level.

influence long-term bond yields depends on the macroprudential accounting regime and, in particular, on banks' MMA exposure. We view this evidence as suggestive, as the monthly time series of path rate shocks does not afford the "clean" event-study type identification of the PSPP. Despite this caveat, Table 9 suggests that the PSPP results generalize to other settings and policy tools aimed at the longer end of the yield curve.

## 7. Discussion: Macroprudential Accounting Rules, Bank Lending, and Capital Requirements

We now discuss some broader implications of the results derived in the previous sections. First, we propose alternative macroprudential regulations that prevent credit crunches induced by increases in sovereign default spreads without muting the pass-through of monetary policy. Second, we argue that macroprudential accounting rules can affect stabilization tools beyond monetary policy. Specifically, we argue that HCA interacts with the capital requirement regulation.

As a first step, we recast our key result in the context of a very simple framework. Consider a bank with deposits  $d_{t-1}$  and two types of assets: loans  $l_{t-1}$ , and sovereign securities with face value  $g$  purchased at  $t-1$  at price  $p_{t-1}$  and with  $t+1$  maturity. The sovereign securities are valued at price  $p$ , which depends on the macroprudential accounting regime: if the bank uses HCA,  $p = p_{t-1}$ ; if instead the bank uses MMA,  $p = p_t$ . The bank is subject to a risk-weighted capital requirement with zero risk weight on sovereign securities and 100% risk weight on loans:

$$\text{capital}_t \geq \xi l_t,$$

where  $\xi$  is the minimum level of the capital that banks need to maintain, and

$$\text{capital}_t = l_{t-1} + p \times g - d_{t-1}.$$

Our first observation is that the framework generates the results that we have derived—about the role of HCA in the pass through of monetary policy—provided that the capital requirement constraint is binding. In particular, when such a constraint is binding, the growth rate of lending is proportional to the accounting value of the sovereign securities,<sup>45</sup>

$$\frac{l_t - l_{t-1}}{l_{t-1}} \propto pg.$$

Under HCA (i.e.,  $p = p_{t-1}$ ), a change in the market price  $p_t$  induced by movement in yields is not transmitted to regulatory capital and, thus, does not affect bank lending. This explains why European regulators have been using

<sup>45</sup> More precisely, the growth rate of loans is affine linear in  $pg$ .

HCA to shield banks from changes in sovereign default spreads, and why the PSPP had almost no effect on lending in 2015, when HCA was prevalent.

Under MMA, the price at which the securities  $g$  are valued is  $p_t = \frac{1}{1+i_t+s_t}$ , where  $i_t$  is the risk-free rate set by the monetary authority, and  $s_t$  is the sovereign default spread. The growth rate of loans is thus

$$\frac{l_t - l_{t-1}}{l_{t-1}} \propto \frac{1}{1+i_t+s_t} g.$$

A reduction in  $i_t$  increases banks' capital and allows banks to lend more. This captures, for instance, the PSPP in 2019 or forward guidance shocks after 2018, when MMA was prevalent. However, an increase in sovereign default spreads also pass through onto the balance sheet and reduce lending.

Our second remark is that a *hybrid* accounting rule eliminates the trade-off between protecting banks from changes in sovereign spreads and allowing monetary policy to pass through. That is, if banks value sovereign securities at the price  $p_t^{hybrid} = \frac{1}{1+i_t+s_{t-1}}$  that fixes default spreads at a historical level,  $s_{t-1}$ , but uses the current risk-free rate  $i_t$ , the growth rate of lending would be

$$\frac{l_t - l_{t-1}}{l_{t-1}} \propto \frac{1}{1+i_t+s_{t-1}} g.$$

This approach requires banks and regulators to separate the yield of risky sovereign debt into a risk-free component and a default premium component. In practice, this does not appear particularly complicated.

A different approach that produces results identical to those of the hybrid rule uses MMA to value assets (i.e., the market value of capital) together with time-varying capital requirements that depend on market prices and banks' balance sheets to target a particular loan supply  $l_t$ :

$$capital_t^{MMA} \geq \zeta_t^* l_t,$$

where  $\zeta_t^*$  is the ratio of the MMA-based value of capital and the hybrid rule-based value of capital:

$$\zeta_t^* = \frac{capital_t^{MMA}}{capital_t^{hybrid}}.$$

The time-varying capital requirement decreases when the sovereign default spread goes up (i.e., when the MMA-based value of capital goes down), fully offsetting the impact of such an increase and, thus, preventing credit crunches. That is,  $\zeta_t^*$  is a countercyclical time-varying capital requirement that avoids the pitfalls of HCA in relation to the pass-through of monetary policy.

Our last observation is that HCA interacts with capital requirements. This is because our results are rationalized in a setting with a binding capital requirement constraint, and because HCA influences the level of regulatory capital. When the central bank reduces the interest rate, our results have shown

that HCA mutes the pass through onto bank lending. What we observe here is that coupling the central bank intervention with an appropriate reduction in capital requirements can undo the effects of HCA, producing the same lending response that would be obtained under MMA.<sup>46</sup> The flip side of this argument is that the combination of HCA and a *static* capital requirement—frequently used in practice—is equivalent to MMA combined with a tightening of the capital requirement in response to changes in yields. Note that the logic extends to changes in yields driven by any exogenous shock and not just those related to monetary policy. For instance, the logic is the same if we focus on shocks that change sovereign default premiums.

To gauge the magnitude of the link between HCA and capital requirements, we provide a simple quantification building on the results of Section 6; details about the calculations are provided in [Internet Appendix VI](#). Consider a 1-percentage-point reduction in the yield on 10-year bonds, and a bank with (a) the average holdings of sovereign securities in the period 2010–2017—when HCA was prevalent—and (b) a 10% Tier 1 ratio.<sup>47</sup> If MMA applies to all the assets of the bank, the bank would increase lending by 5.41%. However, most assets were valued at HCA at that time, and the bank would have increased lending by only 0.40%. This latter figure could be obtained under MMA if the change in yields is coupled with an 8.32% tightening of the capital requirement. This is a large effect, as monetary cycles might result in changes in yields that are three or four times bigger than what we have considered here. In essence, HCA mutes the monetary policy efforts to stimulate the economy—as well as the effects of any reduction in yields—by raising the effective capital requirement that is applied to the market value of banks.

## 8. Conclusion

This paper has used Italian credit register and granular supervisory data to study the role of macroprudential accounting regulation in affecting the pass-through of monetary policy onto banks' loan supply to firms. Our main result is that macroprudential accounting regulation that uses historical cost to compute regulatory capital mutes the transmission of monetary policy onto bank lending, and it arises because the use of historical cost limits the recapitalization of banks and, more precisely, the pass-through of monetary policy onto *regulatory* capital. While our focus is on the ECB's 2015 and 2019

<sup>46</sup> An implicit assumption here is that banks do not issue equity nor do they change their dividend policy in response to the change in capital requirement, which is reasonable when focusing on short-run dynamics. We also abstract from feedback between capital requirements and banks' lending decisions, such as those highlighted by [Bahaj and Malherbe \(2020\)](#). Accounting for such feedback would not change the general message that a change in capital requirement can reestablish the aggregate lending response that would be obtained under MMA.

<sup>47</sup> Note that such a bank is weakly capitalized, as the 10% figure corresponds approximately to the 25th percentile of the Tier 1 ratio distribution in the period we consider. We focus on a weakly capitalized bank because we saw that the effects of the macroprudential accounting regime are stronger for such type of banks, and because the capital requirement constraint is more likely to be binding for such a bank.

announcements of the PSPP—the ECB’s largest QE program—our results extend to other monetary policy shocks to the long end of the yield curve.

To place these results in context, we have noted that a large literature has generally reported mixed results on the real effects of QE with little clarity on the factors behind the wide range of estimates. A related and mostly theoretical literature has also wrestled with whether monetary and macroprudential policies are complements or substitutes. This latter literature has generally focused on macroprudential tools in the form of capital and liquidity regulations. Separately, the literature on accounting itself has mainly focused on the trade-off between market discipline and the costs of excess volatility and contagion. Thus, ours is the first paper, to our knowledge, to use micro data and a plausible identification strategy to study how macroprudential accounting rules can sharply mediate the real effects of yield curve management policies on firms’ access to credit. Our results help explain why estimates of the effect of QE might vary across studies. More generally, our results show that macroprudential accounting policies can directly interfere with other macro-stabilization tools. Besides their effect on limiting monetary policy, we have highlighted the link between macroprudential accounting policies and capital requirements, and we have suggested policies that can achieve the stabilization benefits of macroprudential accounting rules while allowing the pass-through of central bank interventions.

Beyond its policy implications, this paper opens up several directions for future research. Our results suggest that analyses of the trade-off between historical cost and mark-to-market accounting should also incorporate the impact of these accounting regimes on other major macro-stabilization policies, such as monetary policy and capital requirements. Future research can further examine the interaction between these policies as well as others, such as the post-2010 liquidity policy. Our evidence can also help researchers develop more realistic quantitative models of banking regulation and monetary policy that account for the macroprudential accounting regulations.

**Code Availability:** The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/Z1Q5HU>

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